








Modelling Financial Markets

Chapter 8: Backtesting and Evaluating Risk Forecasts



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Opening: Can a Risk Forecast Be Proven Wrong?

> *"The elicitability of a functional opens the possibility to compare competing forecasts and to rank them in terms of their realized scores."*
>
> — *Fissler & Ziegel (2016)*

- GARCH-EVT (GARCH volatility with an extreme value theory tail): the VaR 1% is breached on 1.15% of days on the S&P 500 and on 0.97% on the BET (target 1%)
  - is a rate close to 1% enough to call a model good?
- With $T = 250$ days and a true breach rate of 2%, the Kupiec POF test (proportion of failures) rejects only 24% of the time
  - a doubled breach rate goes unrejected in three years out of four
- The dynamic quantile (DQ) test rejects every S&P 500 model, filtered historical simulation (FHS) included
- ES (Expected Shortfall) cannot be ranked on its own, but the pair (VaR, ES) can, with the consistent FZ0 score
  - FHS has the lowest mean FZ0 loss on the S&P 500 and the BET

▸ Central chart: Power of the POF Test

How Can We Tell Whether a Risk Forecast Is Useful?

- **Calibration**
  - is a VaR (Value-at-Risk) or ES (Expected Shortfall) forecast right, day after day?
  - can a backtest tell?
- **Comparison**
  - which of several forecasts is better?
  - is the gap significant?
- **Decision**: what can we conclude when the tail holds only a few days a year?

Worked exercises: Seminar 8, A1 (Kupiec), A2 (Christoffersen), A4 (Z_2), B2 (FZ0 and DM); readings and derivations: ▸ Appendix

What You Should Be Able to Do After This Chapter

- derive the local power of the POF test and the minimiser of the pinball loss
- compute the tests of the number and the timing of VaR breaches, with the size and power of each test at your T
- compute ES backtests: exceedance residuals, Z_2 by simulation and statistics built on the PIT (probability integral transform)
- interpret rankings of (VaR, ES) forecasts by a consistent score, DM tests and the MCS
  - DM: Diebold–Mariano
  - MCS: model confidence set
- interpret what a distribution-free (conformal) VaR does and does not guarantee
- Detailed examples and advanced topics are in Further Topics, before the Appendix
  - the footer button leads there; the Back button returns to the starting slide
- Further Topics and Appendix: reading for familiarity

What Do We Test? Losses and VaR

- **Backtesting**: compare risk forecasts made with the information $\mathcal{F}_{t-1}$ with the outcomes that followed
  - $\mathcal{F}_{t-1}$: everything known at the close of day $t-1$
- Daily log return: $X_t = r_t = \ln(P_t/P_{t-1})$, where P_t is the closing price on day t
- Daily **loss**: $L_t = -r_t$ (a positive value means money lost)
- **VaR**: the loss exceeded with probability α, minus the α-quantile q_α of the return (Chapter 7)
  - $\mathrm{VaR}_\alpha(X) = -\inf\{x : \ P(X \le x) > \alpha\} = -q_\alpha(X)$
  - equivalently, the $(1-\alpha)$-quantile of the loss L_t
  - e.g. VaR 1% = 3%: the loss should exceed 3% on about 1 day in 100

What Do We Test? Expected Shortfall

- **ES**: the average of the quantiles q_u over all tail levels $u \leq \alpha$, i.e. the mean loss beyond VaR
  - $\mathrm{ES}_\alpha(X) = -\frac{1}{\alpha}\int_0^\alpha q_u(X)\,du = E\left[L_t \mid L_t > \mathrm{VaR}_\alpha\right]$ (continuous case)
  - e.g. Student-t losses with $\nu = 4$, standard deviation 1% and $\alpha = 2.5\%$: VaR $= 1.96\%$, ES $= 2.82\%$
  - always $\mathrm{ES} \geq \mathrm{VaR}$
- Why these levels?
  - VaR 1% is the Basel backtesting level (1996)
  - ES 2.5% is the FRTB capital measure (2019)
- Under the Normal distribution, ES 2.5% $= 2.338\,\sigma$ and VaR 1% $= 2.326\,\sigma$
  - ES 2.5% keeps about the same prudence, but also accounts for the depth of the tail

Breaches and Conditional Calibration

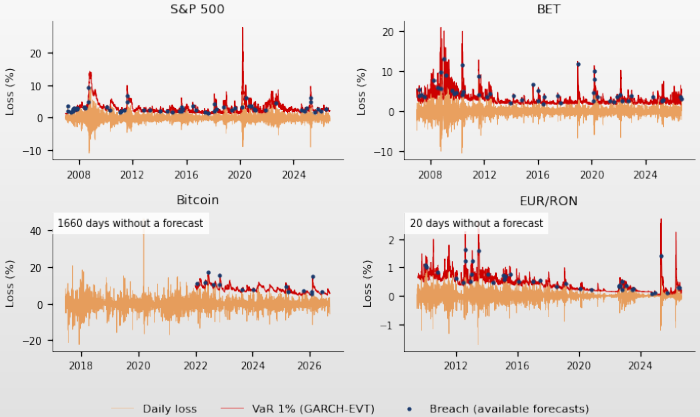
- A **breach** (exception): $I_t = 1\{L_t > \mathrm{VaR}_t\}$, where VaR_t is the forecast made on day $t-1$ for day t
- One-day forecasts: 10-day forecasts would overlap and produce dependent breaches
- **Conditional calibration** (Christoffersen (1998)): $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$ for all t
  - so the I_t are i.i.d. Bernoulli(α)
- Two testable consequences
  - **Unconditional coverage**: about αT breaches in T days
  - **Independence**: breaches do not cluster

Six Ways to Forecast Tail Losses

Model	One-day VaR and ES from
HS	historical simulation: the empirical quantile of order $1-\alpha$ of the last 1000 losses
Normal	the Normal distribution with the window mean and standard deviation
Student-t	a Student-t fitted to the window (location, scale, degrees of freedom ν)
GARCH-t	GARCH (generalised autoregressive conditional heteroskedasticity, Chapter 5): $\sigma_t^2 = \omega + a\,\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2$, with $\varepsilon_{t-1} = r_{t-1} - \mu$; a: reaction to yesterday's shock; b: persistence; Student-t shocks
FHS	filtered HS: GARCH volatility $\times$ empirical quantiles of $z_t = \varepsilon_t/\sigma_t$
GARCH-EVT	EVT (extreme value theory) tail: GARCH volatility $\times$ a GPD (generalised Pareto) tail fitted to the largest 10% of standardised losses (100 exceedances out of 1 000) (McNeil & Frey (2000))

- **Unconditional** (HS, Normal, Student-t): one distribution for the whole 1000-day window
  - every day in the window has the same weight, so the forecast reacts slowly to a volatility burst
- **Conditional** or **filtered** (GARCH-t, FHS, GARCH-EVT): a scale σ_t updated daily by GARCH $\times$ a standardised tail

Four Markets, One Risk Model (1/2)



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Four Markets, One Risk Model (2/2)

- Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ($L_t > \mathrm{VaR}_t$)
- The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- GARCH-EVT VaR 1% breach rate: 1.15% on the S&P 500 and 0.97% on BET (target: 1%)
- On Bitcoin and EUR/RON, the VaR line starts after the withheld GARCH forecasts (Further Topics)
- EUR/RON has a much smaller scale: a managed exchange rate, long calm spells, rare jumps
- What the result supports: description
- Question for the chapter: is a rate close to 1% enough to call a model good?

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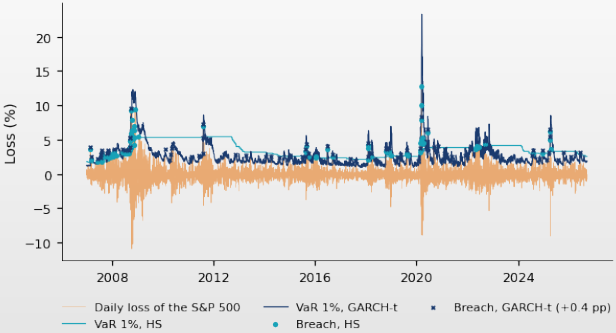
What a Backtest Can and Cannot Identify

- Each backtest checks one implication of conditional calibration
- Non-rejection is not validation
  - it is informative only if the test has power against the relevant alternatives at the available T
- H_0 is a **joint** hypothesis on the whole forecasting procedure
  - distribution, volatility dynamics, estimated parameters, window length and refit rule
- A rejection does not say which component failed
- One outcome per day: evidence accumulates only over many days
- The textbook χ^2 limits treat VaR_t as known
  - the effect of estimation: backtests under estimation risk

Does VaR Have the Right Number of Breaches, at the Right Times?

- Kupiec: the number of breaches
- Christoffersen: first-order memory
- Duration tests: the spacing of breaches
- DQ (Dynamic Quantile): any predictor known yesterday
- For each test: the null, the statistic, its distribution under H_0, and size and power at the available T
- We start with two S&P 500 forecasts with almost the same number of breaches

S&P 500: Two VaR Forecasts, 2007–2026



- Chart: VaR 1% of HS (turquoise) and GARCH-t (blue)
  - daily losses (orange) and each model's breaches (markers)
- HS is a step function: it moves only when extreme days enter or leave the 1 000-day window
- GARCH-t rises from the day after a volatility shock
- Breaches: 83 for HS and 88 for GARCH-t, in 4 959 days
- HS breaches come in bursts (2008, 2020), so **when** they occur must be tested too

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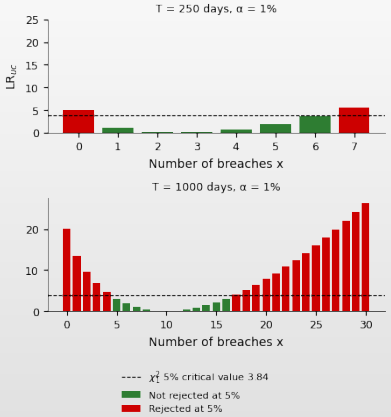
Kupiec (1995): the POF (Proportion of Failures) Test

- What it tests: whether the breach rate equals α (unconditional coverage), whatever their timing
- Data: $x = \sum_t I_t$ breaches in T days; estimated breach rate $\hat{\pi} = x/T$
- H_0: $\pi = \alpha$, against H_1: $\pi \neq \alpha$; π: the true breach probability
- Likelihood ratio statistic (Kupiec (1995)): compares the Binomial likelihood of x under α with that under $\hat{\pi}$
  - $\mathrm{LR}_{uc} = -2 \ln \dfrac{(1-\alpha)^{T-x}\alpha^x}{(1-\hat{\pi})^{T-x}\hat{\pi}^x} \xrightarrow{d} \chi^2_1$
  - $\mathrm{LR}_{uc} \geq 0$ equals 0 when $\hat{\pi} = \alpha$ and grows as $\hat{\pi}$ moves away from α

The POF Test: Decision Rule and an Example

- Reject at 5% if $\mathrm{LR}_{uc} > 3.84$ ($= 1.96^2$, the 95% quantile of χ^2_1)
- The test is two-sided: too few breaches also lead to rejection
- For small T we use the exact Binomial(T, α) distribution of x
- Actual size: the sum of the Binomial probabilities of the rejected counts x
  - with discrete x, no rule has size exactly 5%
- Example: S&P 500, GARCH-t, year 2008: $x = 8$ breaches in $T = 253$ days, against 2.53 expected
- $\mathrm{LR}_{uc} = 7.60$, with $p = 0.0058$ (χ^2_1 approximation)
  - exact two-sided Binomial p-value (doubling the smaller tail): $2P(X \geq 8) = 0.009$
- Both reject at 5%: in 2008 the GARCH-t VaR was breached too often

How Many Breaches Are Acceptable? (1/2)



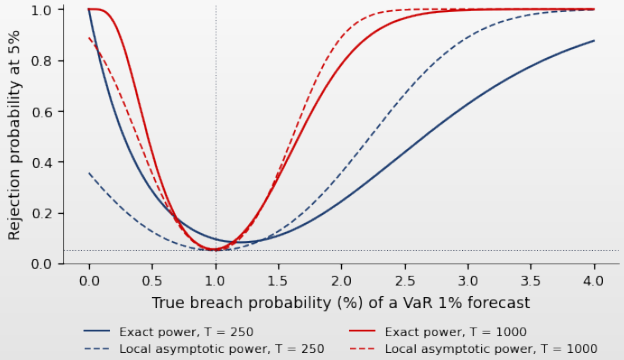
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How Many Breaches Are Acceptable? (2/2)

- LR_{uc} against the number of breaches x, one year (top) and four years (bottom) of VaR 1%
- $T = 250$: the test accepts x from 1 to 6; $T = 1000$: from 5 to 16
- Even $x = 0$ is rejected in one year
  - $\mathrm{LR}_{uc} = -500 \ln 0.99 = 5.03$: a VaR never breached is too conservative
- One year holds about 2.5 informative days for VaR 1%
  - use long samples, two levels ($\alpha = 1\%$ and 2.5%) and several tests

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Power of the POF Test



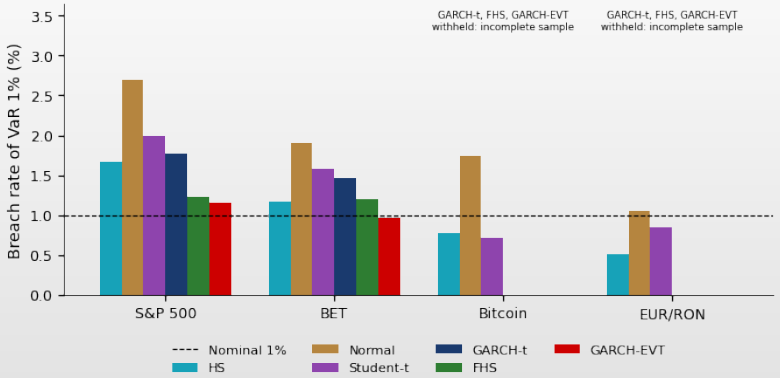
- Axes: true breach rate (horizontal) and rejection probability at 5% (vertical)
- Solid line: exact Binomial calculation; dashed line: local approximation $\chi^2_1(\lambda)$
  - $\lambda = T(\pi - \alpha)^2/(\alpha(1-\alpha))$
- $T = 250$, $\pi = 2\%$: exact power 24%, local power 36%
- So a doubled breach rate goes unrejected in three years out of four
- The rule $\mathrm{LR}_{uc} > 3.84$ has an actual size of 9.5% at $T = 250$; $P(x = 0) = 8.1\%$ alone
- For 50% power: $\pi \approx 2.7\%$ at $T = 250$ and $\pi \approx 1.7\%$ at $T = 1000$

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Power of the POF Test: Local Alternatives

- Local alternatives: the true rate approaches α as T grows, $\pi_T = \alpha + h/\sqrt{T}$
  - h: a fixed distance on the $\sqrt{T}$ scale
- Why: with a fixed $\pi \neq \alpha$ the power tends to 1
  - local alternatives keep it between 0 and 1 and make it computable
- $\mathrm{LR}_{uc} = \dfrac{T(\hat{\pi}-\alpha)^2}{\alpha(1-\alpha)} + o_p(1) \xrightarrow{d} \chi^2_1(\lambda), \qquad \lambda = \dfrac{T(\pi-\alpha)^2}{\alpha(1-\alpha)}$
- λ: the noncentrality parameter; it grows with T and with $(\pi-\alpha)^2$
- Power $= P(\chi^2_1(\lambda) > 3.84)$ grows with λ

Breach Rates Across Models and Markets



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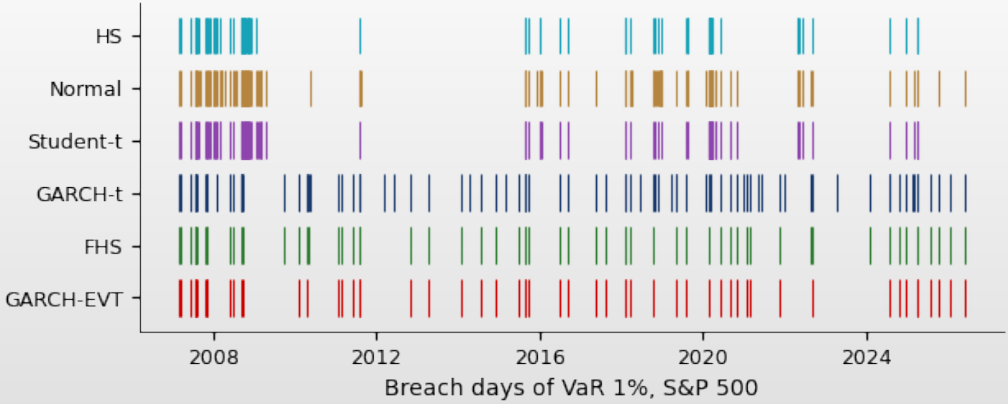
Reading the Breach Rates

- Share of days with $L_t > \mathrm{VaR}_t$ for VaR 1%; the Normal distribution fails except on EUR/RON
  - S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- Same GARCH-t volatility, different tail (S&P 500, BET)
  - GARCH-t 1.77% and 1.46% versus FHS 1.23% and 1.20%
  - evidence against the symmetric t tail on the loss side
- HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high

Christoffersen (1998): Testing Independence

- What it tests: whether today's breach depends on yesterday's
  - the Kupiec test ignores the order of breaches
- Model I_t as a first-order Markov chain with $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- n_{ij}: transitions from state i yesterday to state j today
  - $\hat{\pi}_{01} = \frac{n_{00}}{n_{00}+n_{01}}$, $\hat{\pi}_{11} = \frac{n_{11}}{n_{10}+n_{11}}$
- H_0: $\pi_{01} = \pi_{11} = \pi$ (no memory), with $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- $\mathrm{LR}_{ind} = -2 \ln \frac{(1-\hat{\pi})^{n_{00}+n_{10}} \hat{\pi}^{n_{01}+n_{11}}}{(1-\hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1-\hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi^2_1$
- Numerator: one breach probability; denominator: probabilities that depend on yesterday's state
  - reject at 5% above 3.84
- Conditional coverage: $\mathrm{LR}_{cc} = \mathrm{LR}_{uc} + \mathrm{LR}_{ind} \xrightarrow{d} \chi^2_2$ tests both properties (5% critical value: 5.99)

When Do Breaches Happen? (1/2)



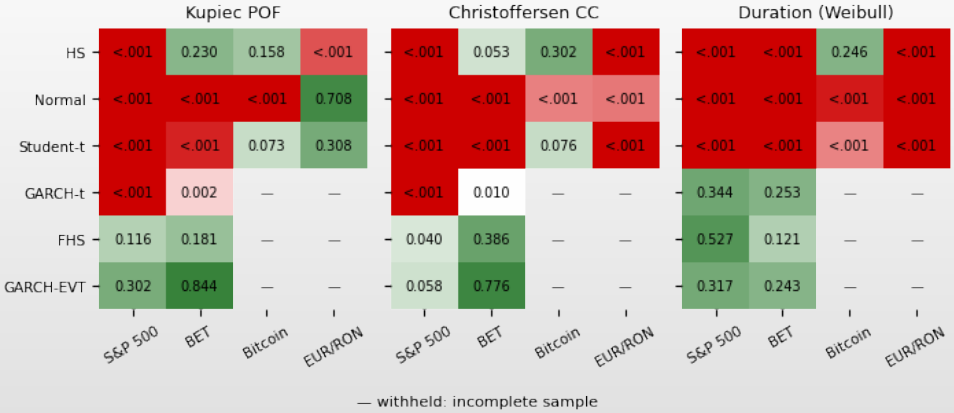
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When Do Breaches Happen? (2/2)

- Each tick marks one breach of VaR 1% on the S&P 500; one row per model
- HS, Normal and Student-t: bursts in 2008, 2015–2016, 2018, 2020, 2022
- Filtered (GARCH) models: no multi-month bursts, yet some back-to-back breaches
- A subsample test can miss clustering: HS, 2007–2010, $n_{00} = 928$, $n_{01} = 38$, $n_{10} = 38$, $n_{11} = 3$
- $\hat{\pi}_{11} = 0.0732$ versus $\hat{\pi}_{01} = 0.0393$, yet $\mathrm{LR}_{ind} = 0.95$, $p = 0.33$
- The reason: $\hat{\pi}_{11}$ rests on only 3 back-to-back pairs

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All VaR Tests at a Glance (1/2)



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All VaR Tests at a Glance (2/2)

- The chart: p-values of the Kupiec POF, conditional coverage (CC) and duration (Weibull) tests, VaR 1%
  - red: rejection, green: no rejection; "—": withheld GARCH forecast
- The DQ test (not in the chart; asymptotic χ^2_6 p-values) rejects every S&P 500 model
  - including FHS ($p < 0.001$) and GARCH-EVT ($p < 0.001$)
  - on BET it does not reject FHS ($p = 0.200$) or GARCH-EVT ($p = 0.279$)
  - hits are predictable from the VaR level and lagged hits, a failure that counting breaches cannot detect
- Different tests examine different failures: read the pattern, not one p-value
  - the unconditional models fail the duration test almost everywhere
  - GARCH-EVT is not rejected by conditional coverage or duration tests on BET
- What the result supports: comparison of procedures

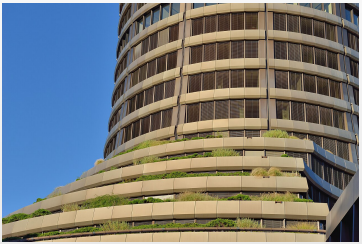
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How Does the Supervisor Test? The Basel Traffic Light

Exceptions	$P(X \leq x)$	Zone
0	8.11%	green
2	54.32%	green
4	89.22%	green
5	95.88%	yellow
7	99.60%	yellow
9	99.97%	yellow
10	99.99%	red

- Banks with internal models compare VaR 1% daily with the actual P&L of the last 250 days (BCBS (1996))
  - P&L: profit and loss
- What it tests: only the number of exceptions, not their size or timing
- $X \sim \text{Binomial}(250, 0.01)$: the exceptions of a correct VaR 1% in 250 days
- $P(X \leq x) = \sum_{k=0}^{x} \binom{250}{k} 0.01^k \, 0.99^{250-k}$: the second column of the table
- Yellow starts where $P(X \leq x) \geq 95\%$, red where $P(X \leq x) \geq 99.99\%$
- Why a count rule: simple, automatic, hard to game

Backtesting Rules in 1996 and 2019

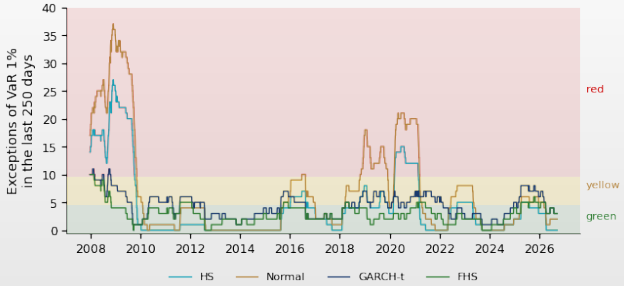


BIS (Bank for International Settlements) tower, Basel: seat of the BCBS secretariat

Photo: Taxiarchos228 (2014); Free Art License; Wikimedia Commons

- BCBS (1996): capital on the 10-day VaR 1%, with multiplier 3
- In 1996: plus 0.40–0.85 in the yellow zone and plus 1 in the red zone
- BCBS (2019) (FRTB): capital on ES 2.5%, but backtesting stays on the VaR 1% of the last 250 days
  - FRTB: Fundamental Review of the Trading Book
- FRTB multiplier, by zone
  - 1.50 in green (0–4 exceptions)
  - 1.70–1.92 in yellow (5–9)
  - 2.00 in red (10 or more)
- A trading desk moves to the standardised approach if, in 12 months, it has
  - more than 12 exceptions of VaR 1%
  - or more than 30 of VaR 2.5%
- IMA (internal models approach) approval also requires a P&L attribution test
- The Basel backtesting rule counts VaR exceptions; ES is not tested directly
  - non-elicitability does not prevent ES backtesting (Acerbi & Szekely (2014), ES section)
  - it concerns the comparison of forecasts (elicitability section)

The Traffic Light Through Time



- VaR 1% exceptions in the last 250 days, S&P 500; bands: green, yellow, red zones
- Share of days in red: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- GARCH-t is yellow 46.5% of the time
- Peak: Normal 37 and HS 27 exceptions, in the 250 days ending 01.12.2008
- Overlapping windows: the penalty lasts a year after a crisis, when capital is already strained
- What the result supports: comparison of procedures

MFM_ch8_traffic_light ▶ Interactive

FRTB Desk Check on the Last 250 Days (Limits 12 and 30)

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	0 / 2	4 / 7	3 / 4	1 / 5
Normal	2 / 2	6 / 9	3 / 5	1 / 4
Student-t	0 / 2	4 / 9	1 / 4	5 / 8
GARCH-t	3 / 6	4 / 9	3 / 5	4 / 5
FHS	3 / 4	4 / 6	4 / 5	3 / 4
GARCH-EVT	3 / 3	2 / 7	4 / 5	3 / 4

- Cells: exceptions of VaR 1% / VaR 2.5% over the last 250 days of each series, up to 18.09.2026
- On these days, GARCH forecasts exist on all four markets
- All models sit comfortably within the limits 12 and 30
- Illustrative only: FRTB applies the limits to desk P&L (actual and hypothetical), not to index returns
- The last year was calm: a green zone today says little about the next crisis

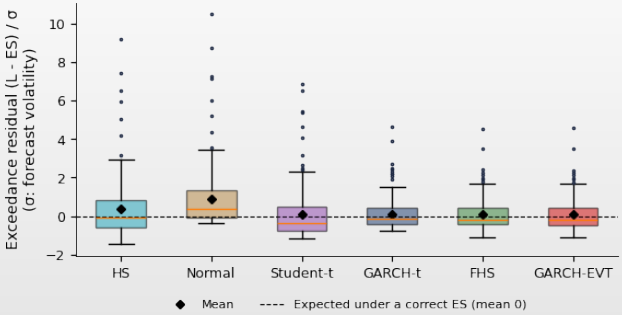
Is ES Forecast Correctly? What Can Be Tested

- ES is the mean loss beyond VaR: the information comes only from about αT tail days
- S&P 500, $\alpha = 2.5\%$: about 124 days out of 4 959; in one year, 6.25 days
- Over one year, the standard error of the tail mean is of order $1/\sqrt{6.25} = 0.40$ standard deviations of the tail losses
- Three families of tests
  - exceedance residuals (McNeil–Frey)
  - the Z_1, Z_2 statistics (Acerbi–Szekely)
  - PIT-based statistics (Du–Escanciano, multinomial tests)
- All need more than the ES forecast: the VaR, the volatility or the whole forecast distribution
  - a test on ES alone: the case study

McNeil and Frey (2000): Exceedance Residuals

- What it tests: whether the mean loss on breach days is the one forecast by ES, on the volatility scale
- Exceedance residual: $e_t = (L_t - \mathrm{ES}_t)/\sigma_t$ on days with $L_t > \mathrm{VaR}_t$ (VaR 2.5%)
  - $\sigma_t > 0$: the volatility forecast at $t-1$
- If VaR **and** ES are right: $E[e_t \mid L_t > \mathrm{VaR}_t, \mathcal{F}_{t-1}] = 0$
  - a wrong VaR changes the conditioning event
- H_0: $E[e_t] = 0$ against H_1: $E[e_t] > 0$ (ES underestimated)
- Statistic: $t = \bar{e}/(s_e/\sqrt{N})$, with $\bar{e}$ and s_e the mean and standard deviation of the N residuals
- p-value: i.i.d. bootstrap of the centred residuals (5 000 replications) (McNeil & Frey (2000))
  - because the residuals are few and heavy-tailed
- The bootstrap assumes independent residuals
  - with dependent residuals, use a block bootstrap

Exceedance Residuals on the S&P 500



- Distribution of $(L_t - \mathrm{ES}_t)/\sigma_t$ on breach days of VaR 2.5%; diamonds: means
- Means: Normal 0.875, HS 0.363, Student-t 0.082, GARCH-t 0.100, FHS 0.104, GARCH-EVT 0.068
- Bootstrap p
  - Normal $<$0.001, HS $<$0.001, Student-t 0.173, GARCH-t 0.025
  - FHS 0.072, GARCH-EVT 0.157
- Normal: tail mean too low by almost one volatility unit
  - the extreme residuals, up to 10 volatilities, matter most for capital

MFM_ch8_es_backtests ▶ Interactive

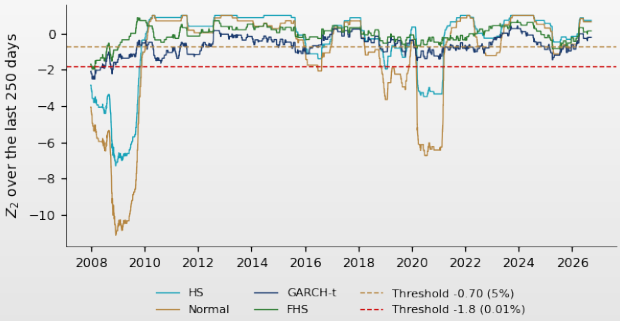
Acerbi and Szekely (2014): Testing ES Directly

- Breach indicator of VaR 2.5%: $I_t = 1\{L_t > \mathrm{VaR}_t^{\alpha}\}$, $\alpha = 0.025$
  - $N_T = \sum_t I_t$: the number of breaches in T days
  - ES_t: the ES forecast for day t
- $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$, conditional on breaches
  - is the average tail loss as large as forecast?
- $Z_2 = 1 - \frac{1}{T\alpha} \sum_t \frac{L_t I_t}{\mathrm{ES}_t}$: tests frequency and size together
- Under H_0 (correct forecast distribution) $E[Z_2] = 0$; underestimated risk gives $Z_2 < 0$ (Acerbi & Szekely (2014))
- For $T = 250$ and many distributions, the 5% and 0.01% critical values are close to -0.70 and -1.8
  - reference thresholds for an ES traffic light

Getting p-Values by Simulation

- The null distribution of Z_2 depends on the forecast distributions F_t, so critical values are simulated
  - F_t: the distribution function of L_t forecast on day $t-1$
- Step 1, repeated $M = 2\,000$ times: for each day draw $U_t \sim \text{Uniform}(0,1)$
  - set $L_t^* = F_t^{-1}(U_t)$ (F_t^{-1}: the quantile function)
- Step 2: compute Z_2^* from the simulated losses, with the same VaR and ES forecasts
  - a calibration with the forecast distributions held fixed and independent U_t: estimation risk and GARCH refitting are ignored
- p-value = share of $Z_2^* \leq$ the observed Z_2; with $M = 2\,000$ the smallest nonzero p-value is 0.0005; no more extreme simulated value means $p < 0.0005$, not $p = 0$
- Only tail draws matter: for HS and FHS we keep the 25 largest losses of each day's window ($25 = 2.5\% \times 1\,000$)
- S&P 500, full sample: $Z_2 = -0.664$ for GARCH-t ($p < 0.001$) and -0.079 for FHS ($p = 0.209$)
- The 5% critical value at $T = 4\,959$ is about -0.150, closer to 0 than -0.70 at $T = 250$

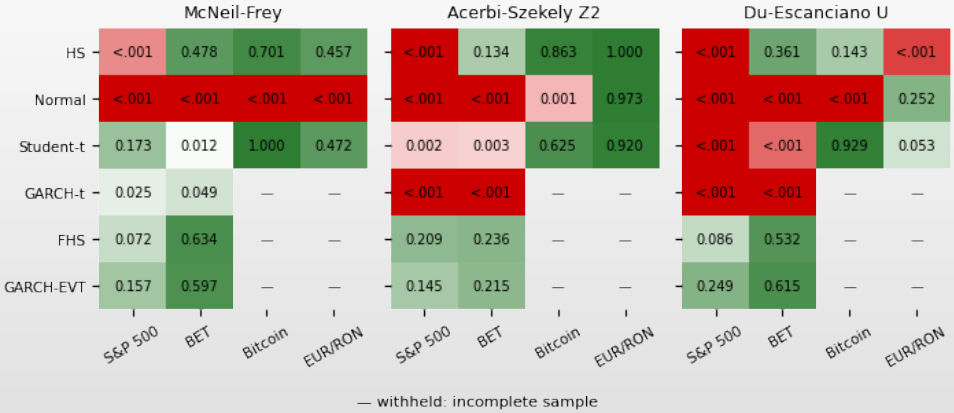
Z_2 Over Rolling 250-Day Windows



- Z_2 for ES 2.5% on the S&P 500, recomputed daily on the last 250 days
  - horizontal lines: the thresholds -0.70 and -1.8
- Minimum: Normal -11.13 (window ending 04.12.2008), HS -7.30 (01.12.2008)
- Minimum: GARCH-t -2.52 (05.02.2008), FHS -1.97 (17.01.2008)
- Windows with $Z_2 < -1.8$: Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%
- GARCH-t is below -0.70 in 44.3% of windows
- Daily checking is repeated testing: false alarms accumulate (sequential monitoring, ▸ Appendix)

MFM_ch8_es_backtests ▶ Interactive

All ES Tests at a Glance (1/2)



MFM_ch8_es_backtests ▶ Interactive

All ES Tests at a Glance (2/2)

- At the 5% significance level, FHS and GARCH-EVT are not rejected on S&P 500 and BET
  - neither by McNeil–Frey, nor by Z_2, nor by Du–Escanciano U, taken separately
- Different nulls and inputs, different decisions
  - Student-t on the S&P 500 has McNeil–Frey $p = 0.173$ but Z_2 $p = 0.002$
- These are calibration tests, not rankings
- GARCH-t is rejected by Z_2 on S&P 500 ($p < 0.001$) and BET ($p < 0.001$)
- Read U with its sign: on EUR/RON, HS has $U = -3.68$ ($p < 0.001$)
  - a rejection because losses stay **shallower** than forecast
  - this matches Z_2 ($p = 1.000$) and its VaR 1% breach rate of 0.51%
- What the result supports: comparison of procedures

MFM_ch8_es_backtests

How Do We Rank Forecasts? Consistent Scores

- A backtest asks whether a model is **acceptable**: several models can pass, or all can fail
- Comparative backtesting asks which model is **better** (Nolde & Ziegel (2017))
- Ranking needs a **scoring function** $S(x, L)$
  - $S(x, L)$: the penalty for reporting x when L occurs
  - compare $\bar{S} = \frac{1}{T}\sum_t S(x_t, L_t)$, lower = better
- S is **consistent** for $\theta(F)$, and θ is **elicitable**, if $\theta(F) = \arg\min_x E_F[S(x, L)]$ (Gneiting (2011))
- $\theta(F)$: the target functional (e.g. VaR) of the loss distribution F; E_F: expectation when $L \sim F$
- Ranking needs the true value to be the **unique** minimiser (strict consistency)

The Pinball Loss and ES

- VaR is elicitable, with the pinball loss $S(v, L) = (1\{L > v\} - \alpha)(L - v)$, where v is the VaR forecast
- If $E|L| < \infty$ and the density $f > 0$ at the quantile:
  $$\frac{\partial}{\partial v} E_F[S(v, L)] = F(v) - (1 - \alpha) = 0 \;\Rightarrow\; v = F^{-1}(1 - \alpha) = \mathrm{VaR}_\alpha$$
- The second derivative is $f(v) > 0$, so the minimum is unique
- ES alone is **not** elicitable: no scoring function of the reported ES e alone is strictly consistent for ES on the relevant class of distributions (Gneiting (2011))
- The reason: the level sets of ES are not convex under mixtures of distributions (Appendix)
- Hence ES is ranked jointly with VaR (next slide)

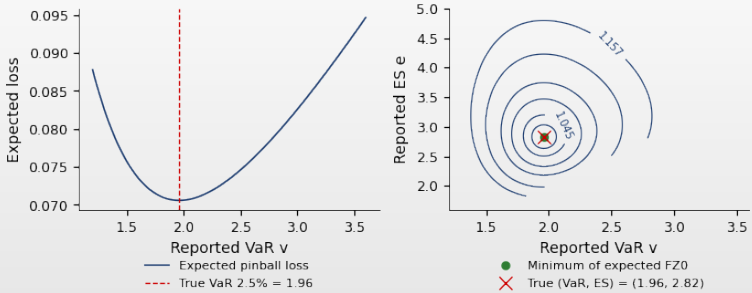
VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: Fissler & Ziegel (2016) characterise the FZ family of consistent scoring functions
- FZ0: the member of the FZ family with zero-homogeneous score differences (Patton, Ziegel & Chen (2019)), for positive losses, $0 < v \leq e$ and $E|L| < \infty$
- $\mathrm{FZ0}(v, e; L) = \dfrac{1\{L > v\}(L - v)}{\alpha\, e} + \dfrac{v}{e} + \ln e - 1$
- v, e: the VaR and ES forecasts (positive losses); L: the realised loss
- The first term penalises breaches, scaled by ES
- The last terms penalise reporting a large ES
- Lower mean FZ0 = better joint VaR and ES forecast
  - $\mathrm{FZ0}(cv, ce; cL) = \mathrm{FZ0}(v, e; L) + \ln c$: differences between models do not depend on the scale of losses

The FZ0 Loss: a One-Day Example

- $\alpha = 2.5\%$, forecasts $v = 3\%$ and $e = 3.9\%$, losses in percent
- Loss $L = 4.5\%$ (breach): $\mathrm{FZ0} = 15.385 + 1.130 = 16.515$
- Loss $L = 1\%$ (no breach): only $v/e + \ln e - 1 = 0.769 + 1.361 - 1 = 1.130$ remains
- In decimals ($v = 0.03$, $e = 0.039$) only $\ln e$ changes
  - each value drops by $\ln 100 = 4.605$
  - to -3.475 without and 11.910 with the breach
- This is why mean FZ0 losses on decimal returns are negative (S&P 500, FHS: -3.551)
  - differences between models are unchanged

Elicitability in Pictures



- Student-t losses with $\nu = 4$ and unit variance, $\alpha = 2.5\%$
- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96
- Right: the expected FZ0 loss is lowest at (1.96, 2.82)
  - the true (VaR, ES) up to the grid
- A forecaster who knows the truth has no incentive to report anything else: this is what makes ranking by average score meaningful

MFM_ch8_elicitability ▶ Interactive

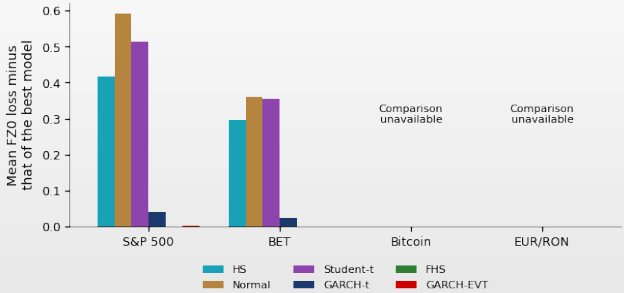
Is the Difference Between Two Forecasts Significant?

- The mean score ranks forecasts, but small gaps can be sampling noise
- Diebold–Mariano: do two forecasts have the same mean loss?
- Giacomini–White: can yesterday's information predict the loss difference?
- Nolde–Ziegel: is the internal model better than a standard model?
- MCS: which set contains the best models with a given probability?
- All use the FZ0 loss of the pair (VaR 2.5%, ES 2.5%)

The Diebold–Mariano Test on FZ0 Losses

- What it tests: whether two forecasts have the same mean loss
- $d_t = \mathrm{FZ0}_{A,t} - \mathrm{FZ0}_{B,t}$; H_0: $E[d_t] = 0$ (equal predictive accuracy), Diebold & Mariano (1995)
- $$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{LRV}}(d_t)/T}} \xrightarrow{d} N(0,1)$$
- $\widehat{\mathrm{LRV}} = \hat{\gamma}_0 + 2\sum_{k=1}^{q}\left(1 - \frac{k}{q+1}\right)\hat{\gamma}_k$: the long-run variance with Newey–West weights (Newey & West (1987))
  - $\hat{\gamma}_k$: autocovariance of d_t at lag k
- Why: tail losses cluster, so d_t is autocorrelated
- Lag rule: $q = \lfloor 4(T/100)^{2/9} \rfloor = 9$ for $T = 4\,959$
- $\mathrm{DM} > 0$: model A has the larger mean loss (worse); $|\mathrm{DM}| > 1.96$: significant at 5%

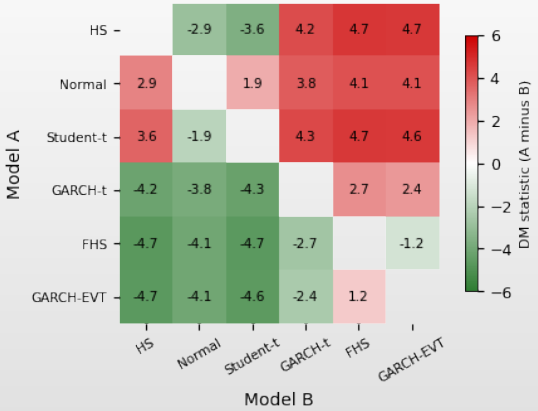
Average FZ0 Loss by Model



- Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market
- On S&P 500 and BET, all six models are compared on the same evaluation days
- Lowest FZ0 loss: FHS on S&P 500 and BET
- The mean FZ0 loss of GARCH-t exceeds that of FHS by 0.040 on the S&P 500
- The size of a gap does not tell whether the data separate the models: we test it

MFM_ch8_comparative ▶ Interactive

Pairwise Diebold–Mariano Tests, S&P 500 (1/2)



MFM_ch8_comparative ▶ Interactive

Pairwise Diebold–Mariano Tests, S&P 500 (2/2)

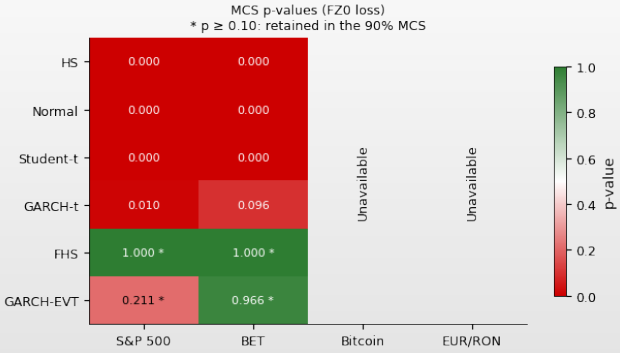
- DM of model A (row) minus B (column) on FZ0 losses; red: A worse, green: A better
- Against FHS
  - HS $\mathrm{DM} = 4.69$, GARCH-t $\mathrm{DM} = 2.66$: both significantly worse
  - GARCH-EVT $\mathrm{DM} = 1.24$: indistinguishable
  - but the conditional Giacomini–White test rejects it against FHS ($p = 0.035$, Further Topics)
- Normal minus HS: $\mathrm{DM} = 2.88$: a slow empirical quantile still beats a thin tail
- Fifteen pairwise tests: a multiple-testing problem
  - which models survive **jointly**? (model confidence set, MCS)

MFM_ch8_comparative

The Model Confidence Set: the Idea

- MCS (Hansen, Lunde & Nason (2011)): the set that contains the best models with asymptotic probability at least $1 - \alpha_{MCS}$
- What it corrects: the 15 pairwise DM tests do not control the family-wise error
  - the MCS compares all models jointly
- The guarantee needs the method's regularity conditions
  - stationary, weakly dependent loss differences
- We use $\alpha_{MCS} = 10\%$: keep models with MCS p-value ≥ 0.10
- A large MCS is an informative result: the data do not separate the models

Which Models Survive?



- MCS p-values on the FZ0 loss of the pair (VaR 2.5%, ES 2.5%)
  - models with $p \geq 0.10$ form the 90% MCS
- S&P 500 and BET: the MCS contains FHS and GARCH-EVT
- GARCH-t is excluded on the S&P 500 ($p = 0.010$) and, narrowly, on BET ($p = 0.096$)
- HS, Normal and Student-t: $p \leq 0.001$ on both markets
- Bitcoin and EUR/RON: no MCS, because GARCH forecasts are withheld (slide on withheld forecasts)
- What the result supports: comparison of procedures

MFM_ch8_comparative ▶ Interactive

Can Coverage Be Guaranteed Without Distributional Assumptions?

- The models so far assume a distribution (Normal, t, GPD) or a stable past (HS)
- **Conformal prediction** (Lei et al. (2018), Angelopoulos & Bates (2021)) turns any forecast into a quantile with a coverage guarantee
- Score: $s_t = L_t/\hat{\sigma}_t$, where $\hat{\sigma}_t$ is the standard deviation of the previous 250 returns
- Calibration set: the last $n = 250$ scores (one trading year), sorted $s_{(1)} \leq \cdots \leq s_{(n)}$
- Split conformal VaR: $\mathrm{VaR}_t = \hat{\sigma}_t \, s_{(k)}$, with $k = \lceil (n+1)(1-\alpha) \rceil$ ($\lceil x \rceil$: the smallest integer $\geq x$)
- For $n = 250$ and $\alpha = 0.01$: $k = 249$, the second-largest score

Split Conformal: the Guarantee and Its Limits

- The guarantee, if the $n+1$ scores are **exchangeable**: $1-\alpha \leq P(L_t \leq \mathrm{VaR}_t) \leq 1-\alpha+\frac{1}{n+1}$ (no ties)
  - exchangeable: the joint law is invariant to permutations
- Here the guarantee is only approximate (extensions: Appendix)
- Volatility clustering breaks exchangeability
- $\hat{\sigma}_t$ comes from overlapping rolling windows, so the scores are not independent
- Genuine split conformal fits the score once, on a separate sample

Adaptive Conformal Inference (ACI)

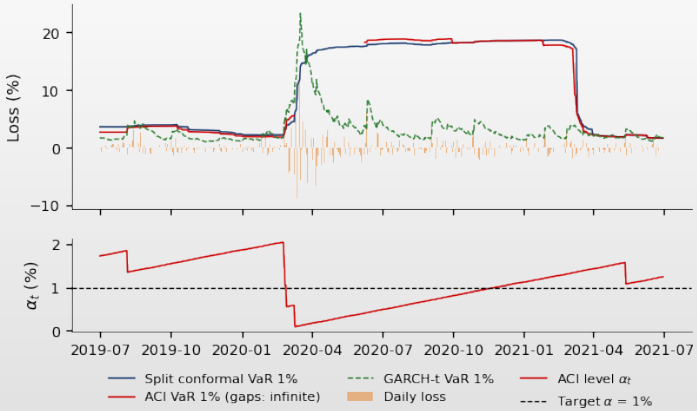


Emmanuel Candès, co-author of adaptive conformal inference

Photo: Cmichel67 (2026); CC BY-SA 4.0; Wikimedia Commons

- Gibbs & Candès (2021): the working level is updated after every day
  - $\alpha_{t+1} = \alpha_t + \gamma\,(\alpha - \mathrm{err}_t)$, $\mathrm{err}_t = 1\{L_t > \mathrm{VaR}_t\}$
  - α_t: the conformal quantile level on day t; $\gamma > 0$: step size; α_1: starting level
- After a breach α_t falls (VaR rises); after calm days it creeps back up
- We use $\gamma = 0.005$: a small step keeps the level stable
  - a large step reacts faster but leads more often to infinite VaR
- The guarantee, for any sequence: $\left|\frac{1}{T}\sum_t \mathrm{err}_t - \alpha\right| \le \frac{\max(\alpha_1, 1-\alpha_1)+\gamma}{\gamma T}$
  - a long-run **average**, not every day or every crisis
- If $\alpha_t < 1/(n+1)$ the required quantile does not exist: VaR $= +\infty$
- If $\alpha_t \ge 1$ the threshold is $-\infty$ and a breach is certain; the guarantee needs both conventions

Conformal VaR Through COVID-19 (1/2)



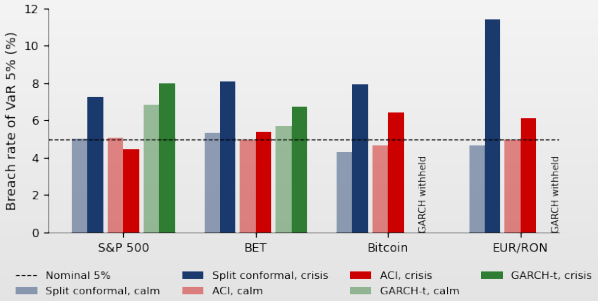
MFM_ch8_conformal ▶ Interactive

Conformal VaR Through COVID-19 (2/2)

- S&P 500, July 2019–June 2021, VaR 1%
  - top: split conformal, ACI and GARCH-t
  - bottom: the ACI level α_t
- From 20.02 to 30.04.2020: 50 days, with 0.5 expected breaches
- Observed breaches: split conformal 6, ACI 4 (8% of days)
- ACI set VaR $= \infty$ on 37 of these days: no new breach, but an infinite VaR carries no information
- The split conformal VaR 1% stays around 18% for a year
  - $\hat{\sigma}_t$ and $s_{(\mathbf{249})}$ use the last 250 days, so they keep the crash until March 2021
- GARCH-t adapts within days: after calm days, the weight of a past shock in σ_t^2 falls by the factor b per day
  - with the parameters estimated on the S&P 500, 1990–2026: $b = 0.892$, half-life of about 6 days
  - the variance forecast reverts on average at rate $a + b = 0.994$: half-life of about 124 days

MFM_ch8_conformal

Coverage in Calm and Crisis Periods



- Chart: VaR 5% breach rate by market and method, on calm and crisis days
- Crisis day, defined in advance: the 20-day volatility exceeds the 90th percentile of the previous 1 000 days
- Why VaR 5%: 713 S&P 500 crisis days give about 36 expected breaches, against 7 for VaR 1%
- Split conformal, all days: S&P 500 5.34%, BET 5.63%, Bitcoin 4.58%, EUR/RON 5.06%
- Split conformal, crisis days: S&P 500 7.29%, BET 8.08%, Bitcoin 7.95%, EUR/RON 11.43%
- ACI, crisis days: S&P 500 4.49%, BET 5.38%
  - for VaR 1%, only with an infinite VaR on 13.4% of days
- Marginal coverage is not crisis coverage
- What the result supports: comparison of procedures

MFM_ch8_conformal ▶ Interactive

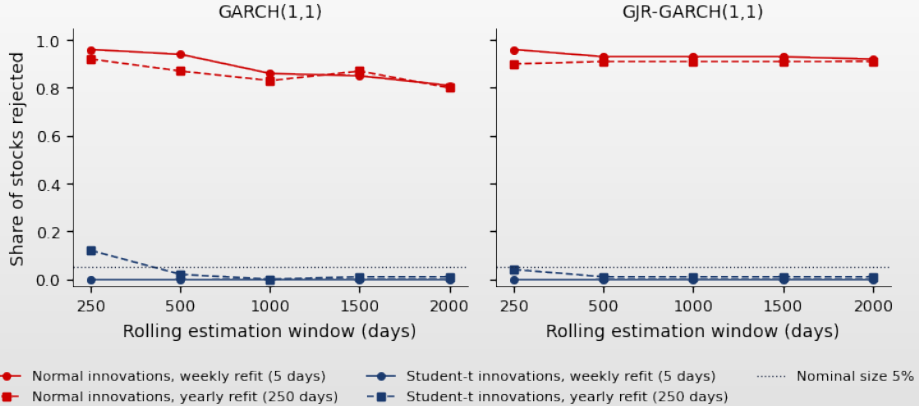
Case Study (1/4): Can ES Be Backtested Using Only ES Forecasts?

- Bayer & Dimitriadis (2022): can ES forecasts be backtested with only the observed returns and these forecasts?
- Relevance: FRTB sets capital from ES 2.5% (the regulator's view)
  - the ES tests above also need VaR, the volatility or the forecast distribution, which supervisors may not receive
  - rejecting such a joint test does not say which forecast is wrong
- The idea: a Mincer–Zarnowitz regression of returns on the ES forecasts
- H_0: intercept 0 and slope 1 (a correct forecast needs no correction)
- The regression can be estimated because (VaR, ES) is jointly elicitable (Fissler & Ziegel (2016))
  - a joint quantile–ES regression (Patton, Ziegel & Chen (2019))

Case Study (2/4): the ES Regression

- Return convention: $\hat{e}_t = -\widehat{\mathrm{ES}}_t < 0$ is the ES forecast on the return scale, with $\tau = \alpha = 2.5\%$
- ES equation: $r_t = \gamma_1 + \gamma_2 \hat{e}_t + u_t^e$, with $\mathrm{ES}_\tau(u_t^e \mid \mathcal{F}_{t-1}) = 0$, so $\gamma_1 + \gamma_2 \hat{e}_t$ is the true conditional ES
- γ_1: intercept; γ_2: slope; correct forecasts: $\gamma_1 = 0$, $\gamma_2 = 1$
- ESR (expected shortfall regression) also needs a quantile equation, because ES is estimated jointly with VaR
- **Auxiliary ESR**: quantile equation on $(1, \hat{v}_t)$, so it also needs VaR forecasts $\hat{v}_t$
- **Strict ESR**: quantile equation on $(1, \hat{e}_t)$, so it needs ES forecasts only
- **Intercept ESR**: $r_t - \hat{e}_t = \gamma_1 + u_t^e$ (slope fixed at 1)
  - one-sided t-test of H_0: $\gamma_1 \geq 0$ against $\gamma_1 < 0$ (ES underestimated)

Case Study (3/4): Tails Matter More Than Refits



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Case Study (4/4): Tails Matter More Than Refits

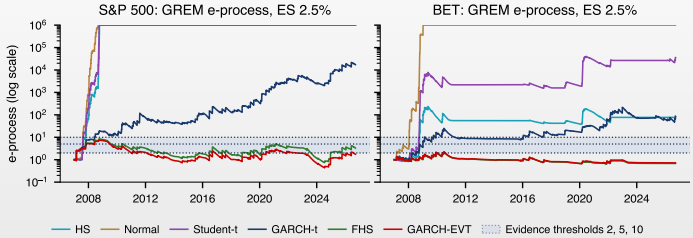
- Source: Bayer & Dimitriadis (2022), Table 3: the 200 largest S&P 500 stocks, 2010–2019
- Axes: estimation window, 250–2 000 days (horizontal)
  - vertical: share of stocks whose ES 2.5% is rejected by the one-sided 5% intercept ESR
- Solid line: refit every 5 days; dashed line: every 250 days (Table 3 also has 21, 62 and 125 days)
- Normal innovations: the rejected share is 0.80–0.96 in all 50 cells
  - 5 windows $\times$ 5 refit frequencies $\times$ GARCH/GJR-GARCH
- GJR-GARCH (Glosten–Jagannathan–Runkle): GARCH with an extra response to negative shocks
- Student-t innovations: at most 0.12 with 250-day windows and at most 0.01 with windows of at least 1 000 days
- Heavy tails and a long window matter most; the refit frequency matters only for short windows
- What the result supports: comparison of procedures
  - the authors note that the classical covariance over-rejects in small samples

MFM_ch8_esr_case

E-Backtesting: Monitoring ES Continuously

- Wang, Wang & Ziegel (2026), *Management Science*: an ES backtest that can be checked every day without losing its validity
- The problem with the ES tests above: a p-value is valid for a sample fixed in advance
  - a supervisor who repeats the test every day and stops at the first rejection has a type I error above 5%
- **E-value**: a nonnegative statistic with mean at most 1 under H_0
  - a large value is evidence against H_0, as a small p-value is
- The paper's claims
  - the test needs only the (VaR, ES) forecasts and the realised losses, not the model that produced them
  - validity needs no asymptotic approximation and holds at any stopping time (anytime-valid)
  - for a fixed loss and VaR, a higher, more prudent ES gives a smaller e-statistic e_t
- The paper's empirical result: NASDAQ Composite, ES 2.5% from AR(1)-GARCH(1,1), from 3 Jan 2005
  - Normal innovations, GREM (growth rate with a mixture) e-process: threshold 2 is reached after 540 trading days, threshold 10 after 713
  - skewed-t innovations with the ES raised by 10%: no threshold reached by 31 Dec 2021

E-Backtesting: ES 2.5% E-Processes on the S&P 500 and BET



- Horizontal: the evaluation days, Jan 2007 – Sep 2026 (S&P 500: 4 959 days; BET: 4 937 days)
  - vertical: the GREM e-process of each model, on a log scale; it starts at 1 (no evidence)
- Band: thresholds 2, 5 and 10; reaching $1/a$ rejects the forecasts with a type I error of at most a (50%, 20%, 10%)
- Above 10^6, the top of the axis: on the S&P 500, Normal, Student-t and HS (4 Sep 2008, 6 Oct 2008, 15 Oct 2008); on BET, Normal (7 Jan 2009)
- FHS and GARCH-EVT stay below 10 on the S&P 500 and below 2.3 on BET

MFM_ch8_ebacktest ▶ Interactive

E-Backtesting: the E-Statistic and the Course Replication

- The e-statistic for ES 2.5%: $e_t = \dfrac{(L_t - z_t)_+}{0.025\,(r_t - z_t)}$, with $(x)_+ = \max(x, 0)$
  - L_t: the loss on day t; z_t: the VaR 2.5% forecast; r_t: the ES 2.5% forecast
  - H_0: for every t, $z_t = \mathrm{VaR}_t$, $r_t \geq \mathrm{ES}_t$ and $r_t > z_t$, with forecasts known at $t-1$
  - under H_0 the conditional mean of e_t is at most 1; a wrong VaR can push it above 1 even with the correct ES
- The e-process: $M_t = \prod_{s \leq t}(1 - \lambda_s + \lambda_s e_s)$, with $\lambda_s \in [0,\ 0.5]$ chosen from the data up to $s-1$ only
  - under H_0, M_t is a nonnegative supermartingale, and Ville's inequality gives $P(\sup_t M_t \geq 1/a) \leq a$
- λ_s maximises the log growth of M_t (the paper's Taylor approximation)
  - GREE (growth rate with empirical e-statistics): from the e-values of the last 500 days
  - GREL (growth rate with empirical losses): from the losses of the last 500 days, at today's forecast
  - GREM (growth rate with a mixture): the average of the two e-processes, the authors' recommended choice
- The course replication, against the paper
  - the chapter's forecasts: six models, rolling window of 1,000 days (paper: AR(1)-GARCH(1,1), 500 days)
  - S&P 500 and BET, 2007–2026 (paper: NASDAQ Composite, 2005–2021); Bitcoin and EUR/RON have withheld GARCH forecasts
  - GREE starts in Jan 2007 with no earlier e-values; the paper had five years of forecasts before 2005

E-Backtesting: Evidence from 2007 On, Except for FHS and GARCH-EVT

Model	S&P 500: max M_t	first day with $M_t \geq 10$	BET: max M_t	first day with $M_t \geq 10$
HS	$1.5 \cdot 10^{9}$	19 Oct 2007	237	10 Oct 2008
Normal	$1.5 \cdot 10^{21}$	9 Aug 2007	$9.3 \cdot 10^{10}$	21 Jan 2008
Student-t	$4.7 \cdot 10^{11}$	15 Aug 2007	$4.1 \cdot 10^{4}$	6 Oct 2008
GARCH-t	$2.2 \cdot 10^{4}$	6 Jun 2008	218	17 Feb 2009
FHS	8.55	—	2.16	—
GARCH-EVT	7.47	—	2.28	—

- S&P 500: Normal reaches threshold 2 on 26 Jul 2007 and threshold 10 on 9 Aug 2007; HS on 19 Oct 2007; GARCH-t on 6 Jun 2008
  - the mean of e_t: 1.89 for GARCH-t and 1.33 for FHS (expected value for a correct ES: 1)
- FHS and GARCH-EVT on the S&P 500: peaks of 8.55 and 7.47 (29 Sep 2008); on BET: 2.16 and 2.28
- HS on BET: Z_2 does not reject on the whole sample ($p = 0.134$), but the e-process reaches 10 on 10 Oct 2008 (peak 237)
- ES raised by 10%: FHS and GARCH-EVT on the S&P 500 stay below 5 (peaks 3.38 and 3.46); GARCH-t on BET stays below 2 (1.79)
  - an empirical result: the bets λ_s are estimated from the past, so the GREM e-process is not guaranteed to be monotone in the ES
- What the result supports: a test of the ES conditional on a correct VaR 2.5%, valid on any day it is read
  - it does not rank the models: that is the job of comparative backtesting and the MCS
- Limits: after a prudent spell, M_t falls and needs time to grow again; thresholds 2 and 5 carry a large type I error (50%, 20%)

An Open Question: Crisis-Conditional Coverage

- Distribution-free VaR guarantees are **marginal**
  - split conformal needs exchangeability
  - ACI controls only the long-run average
  - section "Conformal Prediction for Tail Risk"
- Formal question: is the crisis breach rate still α?
  - with a crisis flag $C_t \in \mathcal{F}_{t-1}$ known ex ante
  - $C_t = 1$ on a crisis day, 0 otherwise
- H_0: $P(L_t > \mathrm{VaR}_t \mid C_t = 1) = \alpha$ against H_1: $P(L_t > \mathrm{VaR}_t \mid C_t = 1) \neq \alpha$
- Tested through the moment $E[C_t(I_t - \alpha)] = 0$, with I_t the breach on day t
  - HAC (heteroskedasticity and autocorrelation consistent) inference or a block bootstrap, because crisis breaches cluster; then the Holm correction
- Stakes: capital is needed exactly on the days when marginal guarantees say least

The Research Loop with an AI Assistant

- The AI assistant: an LLM such as Claude, ChatGPT or Gemini
  - AI: artificial intelligence
  - LLM: large language model
- Literature, with DOIs checked on Crossref
- Replicate a known number first
- Three rival explanations: stale scale, changing tail shape, regime-selection artefact
- Pre-register the tests
- Ask the assistant for a hostile referee report (the prompts: Appendix)

Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Alternative score: the standardised loss $s_t = (L_t + \mu_t)/\sigma_t$, with μ_t, σ_t the GARCH-t forecasts of the mean and volatility
- H_0: $E[C_t(I_t - 0.05)] = 0$, i.e. the crisis breach rate of VaR 5% is 5%
  - t test with a HAC (Newey–West) variance, because crisis breaches cluster
  - Holm correction (a step-down adjustment of the p-values) within each experiment: four markets, and the markets with complete GARCH forecasts (n/a: withheld GARCH forecasts)

Market	Split conformal, full calendar			GARCH scale, common calendar		
	Crisis days	Rate	Holm p	Crisis days	Rate	Holm p
S&P 500	713	7.29%	0.187	613	2.77%	0.003
BET	520	8.08%	0.187	506	3.95%	0.233
Bitcoin	264	7.95%	0.187	n/a	n/a	n/a
EUR/RON	245	11.43%	0.119	n/a	n/a	n/a

ai_discovery_case.py

Mini-Case: Reading the Result

- Two calendars: split conformal on the full calendar of the conformal section
  - the GARCH scale needs 250 calibration days: the S&P 500 has 4 709 days, 613 of them crisis days
  - on these days, the split conformal crisis rate is 7.01%
- Split conformal: crisis rates above 5% on every market, from 7.29% to 11.43%
  - with HAC inference and the Holm correction, no rejection at 5% (smallest Holm p: 0.119, EUR/RON)
- GARCH scale on the S&P 500: 2.77% in crises, below 5% (Holm p 0.003), so too conservative in crises
- GARCH scale on BET: 3.95%, the gap to 5% is not significant (Holm p 0.233)
- Non-rejection does not prove crisis coverage: crisis samples are small and HAC inference is asymptotic
- What the result supports: comparison of procedures

The Practical Decision



National Bank of Romania, Bucharest

Photo: Britchi Mirela (2018); CC BY-SA 4.0; Wikimedia Commons

- **What I choose**: FHS as the main model, GARCH-EVT as the alternative, revisited as evidence accumulates
  - FHS has the lowest mean FZ0 on the S&P 500; on BET, FHS and GARCH-EVT tie to three decimals
- **Under which conditions**
  - FHS and GARCH-EVT pass Z_2 at 5% and stay in the 90% MCS on the S&P 500 and BET (full scorecard: Appendix)
  - The evidence comes from long samples: one year of data has low power
- **What result would make me drop it**
  - A Z_2 rejection at 5% or an exit from the 90% MCS at evaluation dates fixed in advance, adjusted for repetition
  - In daily monitoring: the ES e-process above 20, i.e. the 5% level
  - A rejection of the crisis moment $E[C_t(I_t - \alpha)] = 0$ (HAC inference, Holm)
  - Already against it: DQ rejects every S&P 500 model, FHS included
- A Romanian portfolio (BET and EUR/RON): Seminar 8, C2

Check Your Understanding, then Chapter 9

Retrieval questions

- Why does the rule "$\mathrm{LR}_{uc} > 3.84$" reject a correct VaR 1% in about one year in ten?
- Why can a model with the right number of breaches still fail?
- Why can ES be backtested but not ranked on its own?
- What does ACI guarantee?
- What does ACI not guarantee?

Chapter 9: realised volatility

- Measure daily variance from intraday returns
- Evaluate volatility forecasts with robust scores such as QLIKE (quasi-likelihood loss) and the MCS

Numeric Self-Checks

1. Kupiec: $x = 3$ breaches of VaR 1% in $T = 500$ days; compute $\hat{\pi}$ and LR_{uc}, then decide at 5%
2. FZ0: one day, $\alpha = 5\%$, $v = 2\%$, $e = 2.5\%$
   - compute the score for $L = 3\%$ and for $L = 1\%$, in percent and in decimals
3. Local power: $T = 1000$, $\alpha = 1\%$, true $\pi = 2\%$
   - compute λ and the power, then compare with the dashed curve of the power chart
4. Z_1 and Z_2: $T = 250$, $\alpha = 2.5\%$, five breaches with $L_t/\mathrm{ES}_t = 1.1,\ 0.9,\ 1.3,\ 1.0,\ 1.2$
   - compute both and explain their signs

Answers with intermediate steps: Appendix. Worked exercises on the course data: Seminar 8, A1, A4, B2

Which Data? Four Markets

- S&P 500: a developed, liquid equity market
- BET (Bucharest Stock Exchange index): an emerging, less liquid market
- Bitcoin: trades every day and has very heavy tails
- EUR/RON: the BNR (National Bank of Romania) reference rate, a managed float
- Why these: four different volatility and tail regimes, on which the same tests can behave differently
- S&P 500, BET and Bitcoin: daily closes
  - each series keeps its own trading calendar

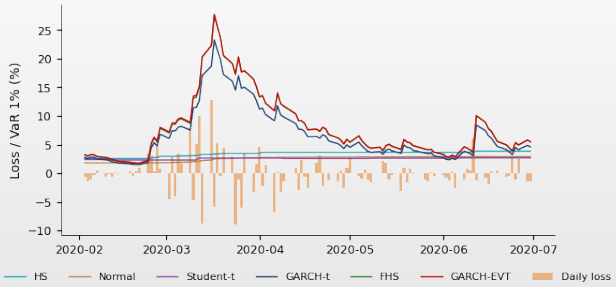
How Are Forecasts Evaluated? The Rolling Window

- Rolling window of 1 000 observations (about four years of equity trading)
  - enough data for GARCH and 100 exceedances for the GPD tail
- Each forecast uses data up to the previous day only (no look-ahead bias)
- GARCH, Student-t and EVT parameters are refitted every 20 days (about one trading month)
  - the GARCH volatility is updated daily
- Evaluation starts on 01.01.2007 or, for a shorter series, after its first 1 000 observations
  - last day: 18.09.2026
- S&P 500 and BET (from 2007): 4 959 and 4 937 evaluation days
- EUR/RON (rate from July 2005, so evaluated from 2009): 4 352 days
- Bitcoin (data from September 2014, so evaluated from 2017): 3 384 days

GARCH Forecasts Withheld on Bitcoin and EUR/RON

- A GARCH(1,1)-t fit is **admissible** if the optimiser converged and $\omega > 0$, $a, b \geq 0$, $\nu > 2$, $a + b < 1$
  - with a numerical margin of 10^{-8}
  - i.e. the unconditional variance $\omega/(1 - a - b)$ is finite
- At an inadmissible refit we keep the last admissible parameters
  - the volatility is still filtered on current data
- With no earlier admissible fit, the GARCH-t, FHS and GARCH-EVT forecasts are **withheld**
  - neither computed nor imputed
- Bitcoin: the first 1 660 of 3 384 evaluation days (the first 83 refits all have $a + b \geq 1 - 10^{-8}$)
- EUR/RON: the first 20 of 4 352 evaluation days (the first refit)
- Full-sample GARCH results and the MCS are reported only on the S&P 500 and BET
  - the MCS needs all six models
  - "n/a" in tables, "—" in charts

March 2020: The Stress Test Nobody Planned



- S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models
- Largest loss: 12.77%, on 16.03.2020
- A correct VaR 1% has 1.04 expected breaches in these 104 days
- Breaches, unconditional models: HS 12, Normal 17, Student-t 17
- Breaches, conditional models: GARCH-t 3, FHS 2, GARCH-EVT 2
- A time-varying scale mattered more than the tail: Student-t (heavy tail, fixed scale) has 17 breaches

MFM_ch8_var_forecasts

Local Power by Hand

- $\chi^2_1(\lambda)$ is the law of $(Z+\sqrt{\lambda})^2$ with $Z \sim N(0,1)$, and $3.84 = 1.96^2$, so
  - power $= P(|Z+\sqrt{\lambda}| > 1.96) = \Phi(\sqrt{\lambda} - 1.96) + \Phi(-\sqrt{\lambda} - 1.96)$, Φ: the $N(0,1)$ distribution function
- $T = 250$, $\pi = 2\%$: $\lambda = 2.525$, $\sqrt{\lambda} = 1.589$
  - power $= \Phi(-0.371) + \Phi(-3.549) = 0.355 + 0.0002 = 0.356$: the 36% of the dashed curve
- The exact power is lower, 24%: the test accepts x from 1 to 6, an event of probability 75.7% when $\pi = 2\%$
  - with 2.5 expected breaches a year, the count is too discrete and too skewed for the Normal approximation behind $\chi^2_1(\lambda)$

Independence on the Full S&P 500 Sample

- p-values of LR_{ind}, VaR 1%: HS, Normal and Student-t <0.001
  - GARCH-EVT 0.031; FHS 0.046; GARCH-t 0.095
- Independence is rejected at 5% for every model except GARCH-t
- After a Holm correction for the six tests, only HS, Normal and Student-t stay rejected
- GARCH-EVT: a breach follows a breach with probability 0.053, against 0.011 after a calm day
- The joint CC test does not reject GARCH-EVT ($p = 0.058$): its correct coverage part dilutes the signal

MFM_ch8_coverage_tests

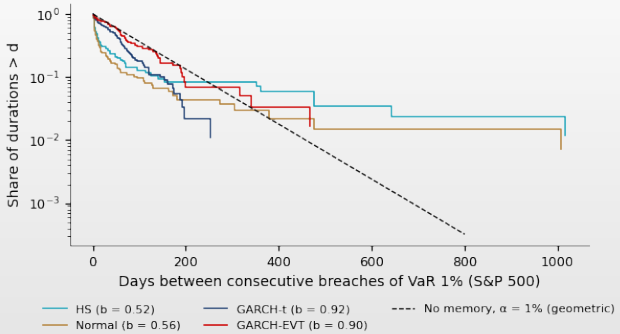
Duration-Based Tests

- What it tests: whether the gaps between breaches are those of a memoryless process
  - it uses every gap, not only back-to-back pairs
- Duration D_i: the number of days between breach $i-1$ and breach i
- For a correct VaR, D_i is geometric with mean $1/\alpha$ (100 days for $\alpha = 1\%$)
- Christoffersen & Pelletier (2004): approximate the discrete durations by the Weibull density $f(D; a, b) = a^b \, b \, D^{b-1} e^{-(aD)^b}$
- $a > 0$: rate (inverse scale); $b > 0$: shape
- $b = 1$: the exponential, memoryless case
- $b < 1$: short durations are too frequent, i.e. breaches cluster

The Duration Test: the Statistic

- H_0: $b = 1$ with a free, against H_1: $b \neq 1$
- $\mathrm{LR}_{dur} = 2\left[\ell(\hat{a}, \hat{b}) - \ell(\tilde{a}, 1)\right] \xrightarrow{d} \chi^2_1$, where $\tilde{a}$ maximises the likelihood with $b = 1$
- $\ell(a, b) = \sum_i [C_i \ln S(D_i) + (1 - C_i) \ln f(D_i)]$, with $S(D) = e^{-(aD)^b}$ the survival function
- $C_i = 1$ for the first and last durations, which are censored
  - the first does not start at a breach, the last does not end at one
- The test does not check the mean $1/\alpha$: coverage is left to the Kupiec test
- At $T \geq 1\,000$ it detects clustering better than the conditional coverage test
  - at $T = 250$ it does not (the Monte Carlo at the end of the section)

How Long Between Breaches?



- Axes: duration d in days (horizontal) and the share of durations longer than d (vertical)
  - the vertical axis is on a log scale
- Dotted line: memoryless (geometric), $\alpha = 1\%$
- HS: $b = 0.52$, $p < 0.001$; median duration 7 days
- Normal: $b = 0.56$
- GARCH-t: $b = 0.92$, $p = 0.344$; median duration 45 days
- GARCH-EVT: $b = 0.90$, $p = 0.317$
- The long HS tail: years without breaches after a crisis leaves a high VaR in the window

MFM_ch8_coverage_tests

Regression-Based Backtests: the DQ Test

- What it tests: whether breaches can be predicted from any information known yesterday
  - not only from yesterday's breach
- Under conditional calibration $\mathrm{Hit}_t = I_t - \alpha$ is a martingale difference
  - $E[\mathrm{Hit}_t \mid \mathcal{F}_{t-1}] = 0$, $\mathrm{Var}(\mathrm{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1-\alpha)$
- Regress $\mathrm{Hit}_t = X_t\beta + u_t$ with $X_t = (1, \mathrm{Hit}_{t-1}, \ldots, \mathrm{Hit}_{t-4}, \mathrm{VaR}_t)$ and test $\beta = 0$ (Engle & Manganelli (2004))
- $\mathrm{DQ} = \dfrac{\hat{\beta}' X' X \hat{\beta}}{\alpha(1-\alpha)} = \dfrac{\mathrm{Hit}' X (X'X)^{-1} X' \mathrm{Hit}}{\alpha(1-\alpha)} \xrightarrow{d} \chi^2_6$
- X: the $T \times 6$ regressor matrix; Hit: the vector of centred hits
  - $\hat{\beta}$: the OLS (ordinary least squares) estimator
  - u_t: the error
- DQ: the explained sum of squares, scaled by the null variance

The DQ Test: Special Cases and Size

- With $X_t = 1$: the score form of the Kupiec test, $T(\hat{\pi} - \alpha)^2/(\alpha(1-\alpha))$
- With $X_t = (1, \mathrm{Hit}_{t-1})$: a 2-degree-of-freedom analogue of LR_{cc}
  - asymptotically equivalent but not numerically equal
- Testing only the lag coefficient corresponds to LR_{ind}
- What it adds: power against memory beyond lag 1 and against breaches that depend on the VaR level
- Limitation: the χ^2_6 approximation over-rejects a correct model
  - the Monte Carlo at the end of the section
- The DQ p-values in this chapter are asymptotic; critical values have to be simulated
  - under the data-generating process of the Monte Carlo experiment, the 95% quantile of DQ at $T = 4\,959$ is 14.2, against 12.59 for χ^2_6
  - FHS and GARCH-EVT on the S&P 500: simulated frequencies of 0.6% and 0.5%, against asymptotic p-values of 0.014% and 0.011%
  - a benchmark under this teaching process, not definitive p-values for the empirical data

Size and Power of VaR Backtests: a Monte Carlo

- DGP (data generating process, the simulated truth): GARCH(1,1)-t fitted to the S&P 500 1990–2026
  - $a = 0.102$, $b = 0.892$, $\nu = 6.1$
  - VaR 1%; 1000 paths; rejection rates (%) at 5%

Forecaster	T	POF	Cond. cov.	Duration	DQ
Correct (size)	250	**9.7**	1.1	15.0	8.6
	1000	6.4	3.3	8.0	11.0
	5000	4.9	3.4	4.6	**8.8**
Normal-GARCH (power)	250	**10.6**	4.8	14.8	18.8
	1000	30.2	19.9	8.2	30.2
	5000	83.7	77.2	7.3	70.7
HS, 1000 days (power)	250	51.5	17.0	**12.0**	32.5
	1000	46.4	46.9	68.3	73.8
	5000	57.1	86.6	100.0	99.5

MFM_ch8_size_power

Size and Power of VaR Backtests: How to Read the Table

- Correct: true parameters and σ_t; the rejection rate is the size (target: 5%)
- Normal-GARCH: true σ_t, Normal tail; the rejection rate is power
- HS, 1 000 days: no volatility dynamics; the rejection rate is power
- An undefined duration p-value counts as non-rejection
  - undefined: fewer than 3 durations, in 28% of correct samples at $T = 250$
- Monte Carlo standard error: 0.7 percentage points at 5% and 1.6 at 50%
- In the bold POF cell (9.7%) the standard error is 0.9: the size is clearly above 5%

Size and Power of VaR Backtests: What the Table Shows

- Within a year, a Normal tail on the correct volatility is nearly invisible
  - POF rejects 10.6% of the time, about its size (9.7%)
- HS clustering is rarely detected by the duration test at $T = 250$ (12.0%)
- At $T = 1\,000$ the duration test (68.3%) and DQ (73.8%) detect HS far better than POF (46.4%)
- The conditional coverage test is conservative: size 1.1% at $T = 250$
- DQ over-rejects a correct model even at $T = 5\,000$ (8.8%): its critical values should be simulated
- At $T = 250$ no test has both correct size and high power

Backtests Under Estimation Risk

- Forecasts use $\hat{\theta}$ estimated on R days and the test uses P out-of-sample days
  - the hits $\hat{I}_t = 1\{L_t > \mathrm{VaR}_t(\hat{\theta})\}$ are not Bernoulli(α) even under a correct model
- The correction depends on the forecasting scheme (fixed, rolling, recursive) and grows with P/R (Escanciano & Olmo (2010))
  - expansion in the Appendix
- Fixed scheme, 400 GARCH(1,1)-t paths, $R = 1000$
  - the fit is admissible on 324 of them (simulation results)
- This chapter uses a rolling window ($R = 1000$, refitted every 20 days)
  - a fixed-scheme experiment does not measure the size of these rolling-window tests
- Remedy: reproduce the estimation inside the calibration
  - e.g. a parametric bootstrap that re-estimates the model on every simulated path
  - Seminar 8, B9, also discusses the selection of admissible paths

Du and Escanciano (2017): Cumulative Violations

- What it tests: how deep into the tail losses go, not only whether they breach VaR
  - it uses the forecast distribution through the PIT
- PIT: $u_t = F_t(L_t)$, uniform on $(0,1)$ if the forecast distribution F_t is right
- Cumulative violation $H_t = \frac{1}{\alpha}\left(u_t - (1-\alpha)\right) 1\{u_t > 1-\alpha\}$
  - under H_0: $E[H_t] = \alpha/2$, $\mathrm{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- Unconditional test: $U = \sqrt{T}\,(\bar{H} - \alpha/2)/\sqrt{\alpha(1/3-\alpha/4)} \xrightarrow{d} N(0,1)$, with $\bar{H}$ the sample mean of H_t
- The sign of U
  - $U > 0$: losses deeper than forecast (risk underestimated)
  - $U < 0$: shallower (risk overestimated)
- The heat map in all ES tests at a glance gives the two-sided $p = 2\Phi(-|U|)$
  - the one-sided 5% test rejects for $U > 1.645$
- Conditional test (Du & Escanciano (2017)): the Box–Pierce statistic $C_5 = T\sum_{j=1}^{5} \hat{\rho}_j^2 \xrightarrow{d} \chi^2_5$
  - $\hat{\rho}_j$: the autocorrelations of H_t

Multinomial VaR Backtests: an Implicit ES Test

- ES 2.5% averages the quantiles beyond VaR 2.5%: test several VaR levels at once (Kratz, Lok & McNeil (2018))
- $N = 4$ levels $\alpha_j = 2.5\%\,(1 - (j-1)/4)$: 2.5%, 1.875%, 1.25% and 0.625%
  - by its PIT u_t each day falls in one of 5 cells
- Pearson statistic: $S = \sum_{j=0}^{4} (O_j - Tp_j)^2/(Tp_j) \xrightarrow{d} \chi^2_4$, with O_j the observed and Tp_j the expected days in cell j
- Cell probabilities: $p_0 = 97.5\%$, $p_1 = \cdots = p_4 = 0.625\%$
  - a large S flags counts far from the expected ones

p-value	S&P 500	BET	Bitcoin	EUR/RON
HS	<0.001	0.725	0.564	0.004
Normal	<0.001	<0.001	<0.001	<0.001
Student-t	<0.001	0.024	0.015	0.172
GARCH-t	<0.001	<0.001	n/a	n/a
FHS	0.003	0.862	n/a	n/a
GARCH-EVT	0.378	0.853	n/a	n/a

- n/a: withheld GARCH result (slide on withheld forecasts)

Multinomial Cell Counts: S&P 500, GARCH-t

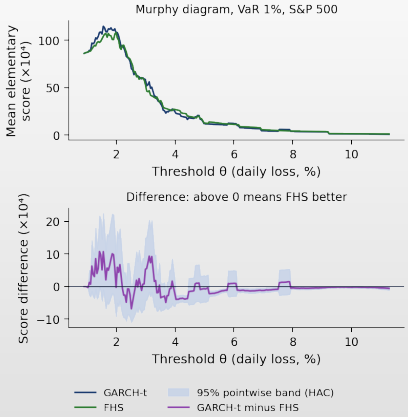
PIT interval	Observed days	Expected days
[0, 0.975]	4761	4 835.02
(0.975, 0.98125]	39	30.99
(0.98125, 0.9875]	46	30.99
(0.9875, 0.99375]	52	30.99
(0.99375, 1]	61	30.99

- The first cell lies outside the upper 2.5% loss tail
- Each of the four tail cells contains more days than expected under calibration

Is the Ranking Robust to the Score? Elementary Scores

- A quantile has infinitely many consistent scores: $S(v, L) = (1\{L \leq v\} - \tau)(g(v) - g(L))$, with g increasing and $\tau = 1 - \alpha$
- Each is a mixture of **elementary scores** (Ehm, Gneiting, Jordan & Krüger (2016))
  - $S_\theta(v, L) = (1\{L < v\} - \tau)\,(1\{\theta < v\} - 1\{\theta < L\})$
  - $\theta \in \mathbb{R}$: a threshold on the loss scale
- S_θ penalises a forecast only when v and L fall on opposite sides of θ
- The pinball loss is $\int S_\theta \, d\theta$
- A is preferred to B under **every** consistent score if and only if $\bar{S}_\theta(A) \leq \bar{S}_\theta(B)$ for all θ
  - $\bar{S}_\theta$: mean elementary score over T days
- **Murphy diagram**: $\bar{S}_\theta$ against θ
- If the curves cross, the ranking depends on the chosen score

Murphy Diagram: GARCH-t versus FHS (1/2)



MFM_ch8_comparative

Murphy Diagram: GARCH-t versus FHS (2/2)

- S&P 500, VaR 1%, 2007–2026; θ from 0.9% to 11.3%
  - pinball ($\times 10^4$): GARCH-t 3.852, FHS 3.861, a tie under the usual score
- Below $\theta = 5\%$ the curves cross
  - FHS better on 47% of thresholds, band containing zero on 86%
  - above 5% GARCH-t lower on 89%: no dominance
- The 95% bands use HAC standard errors and are pointwise, not a simultaneous test of dominance
  - HAC: heteroskedasticity- and autocorrelation-consistent
- State the score before looking at the results

MFM_ch8_comparative

Giacomini–White: Why Compare Methods, Not Models

- DM targets the population accuracy of **models**
- With rolling 1 000-day windows and refits we compare **forecasting methods**
  - estimation error does not vanish as T grows
- Giacomini & White (2006): H_0: $E[d_t \mid \mathcal{F}_{t-1}] = 0$ for $d_t = \mathrm{FZ0}_{A,t} - \mathrm{FZ0}_{B,t}$
- The test is valid for nested models too, with finite rolling windows (not with recursive ones)
- What it adds: it tests whether the loss difference can be predicted from information known at $t-1$
  - not only whether the mean gap is zero

Giacomini–White: Statistic and Results

- Test function $h_{t-1} = (1, d_{t-1})'$, known at $t-1$; $Z_t = h_{t-1} d_t$
- $\mathrm{GW} = T\,\bar{Z}'\hat{\Omega}^{-1}\bar{Z} \xrightarrow{d} \chi^2_2$, $\hat{\Omega} = T^{-1}\sum_t Z_t Z_t'$ (one-step forecasts)
- $\bar{Z}$: the mean of Z_t; 2 degrees of freedom, one per instrument; $h_{t-1} = 1$ gives the DM form
- S&P 500, against FHS: GARCH-EVT $\mathrm{GW} = 6.71$ ($p = 0.035$), GARCH-t 7.60 ($p = 0.022$), HS 53.16 ($p < 0.001$)
- GARCH-EVT, tied under DM, is rejected conditionally
  - equal conditional predictive ability is rejected for the instruments h_{t-1}
- The rejection alone does not show when FHS scores lower
- Which instrument drives the rejection shows only when each is tested separately

Three-Zone Comparative Backtesting (Nolde and Ziegel)

- Compare the internal model A with a standard model S (here HS) by mean FZ0 loss (Nolde & Ziegel (2017))
  - with two one-sided 5% tests
- H_0^-: A is at least as good as S; H_0^+: A is at most as good
- **Green**: reject H_0^+ ($\mathrm{DM} < -1.645$); **red**: reject H_0^- ($\mathrm{DM} > 1.645$); **yellow**: neither
- E.g. $\mathrm{DM} = -1.8$ is green, although $|\mathrm{DM}| < 1.96$
- The burden of proof flips: the bank must show its model is better, not the supervisor that it is worse

DM (A minus HS)	S&P 500	BET	Bitcoin	EUR/RON
GARCH-t	−4.21	−5.25	n/a	n/a
FHS	−4.69	−5.53	n/a	n/a
GARCH-EVT	−4.67	−5.57	n/a	n/a

- Colour gives the statistical zone; n/a: withheld GARCH result (slide on withheld forecasts)

The Model Confidence Set: the Statistic

- For model i in the current set $\mathcal{M}$: $d_{i\cdot} = \bar{L}_i - \frac{1}{|\mathcal{M}|}\sum_{j\in\mathcal{M}} \bar{L}_j$
  - its mean loss relative to the set average
- $t_i = d_{i\cdot}/\widehat{\mathrm{se}}(d_{i\cdot})$ and $T_{\max} = \max_{i\in\mathcal{M}} t_i$; H_0: all models in $\mathcal{M}$ have the same expected loss
- $\widehat{\mathrm{se}}$ and the null distribution of $T_{\max}$ come from a circular block bootstrap with 10-day blocks
  - with 1 000 replications
- If H_0 is rejected, drop the model with the largest t_i and repeat
- MCS p-value of a model: the largest p-value of the tests up to its elimination
- With 1 000 replications the smallest resolvable p-value is about 0.001: a "0.000" in the chart means $p \leq 0.001$

Case Study (1/5): What Is Estimated, Under Which Assumptions

- M-estimator: the parameters minimise the sample mean of an FZ0-type loss
- Asymptotic theory under misspecification (the paper, Theorem 1.4)
  - R packages esback and esreg
- Assumption 1, for strict and intercept ESR
  - the quantile equation is right only if VaR is an affine function of ES
  - e.g. pure scale models
- The covariance estimator used here estimates some components under a location-scale assumption
  - the sandwich covariance under misspecification does not, in general, need this assumption
- Sandwich covariance: $A^{-1}BA^{-1}$; A: curvature of the mean loss; B: variance of its gradient
- Location-scale model: $r_t = \mu_t + \sigma_t z_t$ with i.i.d. standardised shocks z_t

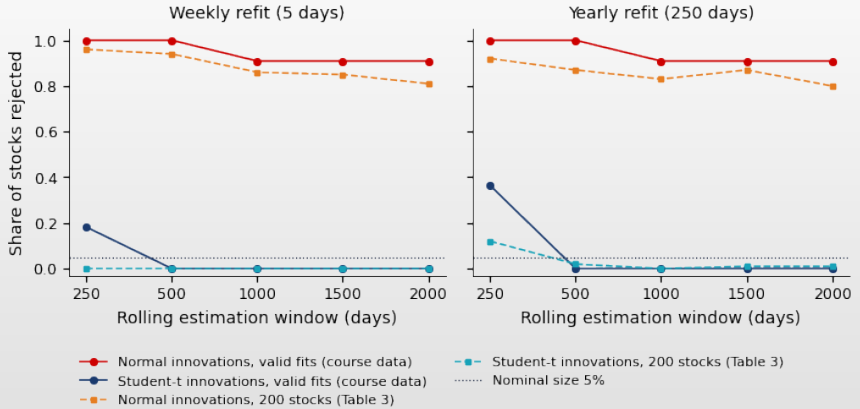
Case Study (2/5): the Paper's Monte Carlo Design

- The paper, Sec. 2: 10 000 replications, T from 250 to 5 000
- EGARCH-t (exponential GARCH, Student-t shocks), calibrated to the S&P 500
- AR(1)-GARCH(1,1): first-order autoregressive mean, GARCH variance
- Two Student-t GAS (generalised autoregressive score) models
- Benchmarks: McNeil & Frey (2000) and the calibration tests of Nolde & Ziegel (2017)

Case Study (3/5): Size and Power in the Paper's Simulations

- Actual size at nominal 5%, EGARCH-t, $T = 250$ and $T = 1\,000$ (Table 1)
  - strict ESR, robust covariance: 0.09 and 0.05
  - strict ESR, classical covariance: 0.24 and 0.11
  - simple conditional calibration test: 0.29 and 0.14
- Power (Fig. 3): exceedance residual tests barely react to two errors
  - wrong volatility dynamics or a wrong level $\tilde{\tau} \neq \tau$
  - ESR detects both
- A bank could report ES at a less extreme level without being rejected by the older tests

Case Study (4/5): Table 3 Replicated on the Course Data



MFM_ch8_esr_case

Case Study (5/5): Table 3 Replicated on the Course Data

- Solid line: 11 large S&P 500 stocks, January 2010 – August 2019 (2 432 days); dashed line: Bayer & Dimitriadis (2022), Table 3
- GARCH(1,1) with constant mean; 25 window–refit combinations (the 5×5 grid)
  - intercept ESR with esback
- Normal innovations: the rejected share is 0.82–1.00 (paper: 0.80–0.96)
- Student-t, windows of at least 1 000 days: no stock rejected (paper: at most 0.01)
- Student-t, 250-day windows: 0.09–0.55 (paper: 0.00–0.12)
- With 11 stocks each rejection adds $1/11 \approx 0.09$, and the standard error of a share reaches $\sqrt{0.25/11} \approx 0.15$, an i.i.d. benchmark: the stocks share market shocks
- The central result replicates
  - the gap at the 250-day window is large even against this standard error

MFM_ch8_esr_case

Further Reading and Resources

- Fissler & Ziegel (2016): VaR and ES are jointly elicitable, the basis of the FZ0 loss
- Nolde & Ziegel (2017): comparative backtests and the traffic light for VaR and ES
- Acerbi & Szekely (2014): model-free ES backtests, including Z_2
- Entropy-based VaR and ES forecasts: Pele, Lazar & Dufour (2017), Pele & Mazurencu-Marinescu-Pele (2019), Pele, Lazar & Mazurencu-Marinescu-Pele (2019); when two ES forecasts can be told apart: Pele & Mazurencu-Marinescu-Pele (2026)
- Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)
- Quantinar: Machine learning in Financial Risk
- Code and data: github.com/danpele/MFM/tree/main/Quantlets/Ch_08
  - S&P 500, BET, Bitcoin: daily closes from the course data
  - EUR/RON: BNR reference rate

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Estimation Risk, the Fixed-Scheme Expansion

- Fixed scheme (Escanciano & Olmo (2010)): $\frac{1}{\sqrt{P}} \sum_t (\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (I_t - \alpha) + \sqrt{P/R}\, D' \sqrt{R}(\hat{\theta} - \theta_0) + o_p(1)$
- First term: hits centred at the true parameters θ_0
  - second: the effect of the estimation error $\hat{\theta} - \theta_0$
- $D = E\left[\partial P(L_t > \mathrm{VaR}_t(\theta) \mid \mathcal{F}_{t-1})/\partial\theta\right]$ at θ_0
  - the sensitivity of the breach probability to the parameters
- The estimation term adds variance of order P/R to $\sqrt{P}(\hat{\pi} - \alpha)$ and vanishes only if $P/R \to 0$
- So the χ^2 critical values are too small when P/R is large
- Assumptions: smooth $\mathrm{VaR}_t(\theta)$, $\sqrt{R}$-consistent $\hat{\theta}$, stationarity
  - rolling and recursive schemes change the form of the correction
- Remedies: the corrected variance or subsampling of Escanciano & Olmo (2010)
  - or a parametric bootstrap that refits the model on every simulated path

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Estimation Risk: Simulation Results

- Fixed scheme: 400 paths, $R = 1000$, Kupiec test at 5%
- Fits on the 400 paths
  - 324 admissible
  - 75 numerically unidentified
  - 1 fails the finite-variance admissibility condition
- We report only the true-parameter rates, on all 400 paths
  - a rate on the 324 admissible paths would be conditional on estimation success
- True parameters: 10.0% at $P = 250$, 5.2% at $P = 1000$, 4.0% at $P = 5000$
- Monte Carlo standard error at a 5% rejection probability: about 1.1 percentage points

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Osband's Principle and Identification Functions

- Osband's principle (Fissler & Ziegel (2016)): consistent scores are built from an identification function V
  - V has $E_F[V(x, L)] = 0$ only at the true value
- VaR: $V(v, L) = 1\{L \leq v\} - (1 - \alpha)$; the pinball loss is its integral in v
- (VaR, ES): $V_1 = \alpha - 1\{L > v\}$ and $V_2 = e - v - \frac{1}{\alpha} 1\{L > v\}(L - v)$
  - their expectations vanish only at the true pair
- The level sets of ES are not convex under mixtures
  - so no identification function of e alone exists (Seminar 8, A8)
- FZ0 is the member of the FZ family whose differences are invariant to the scale of the losses (Patton, Ziegel & Chen (2019))

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Sequential Monitoring of Risk Forecasts

- A regulator checks the forecasts every day
  - repeating a fixed-sample test (e.g. rolling Z_2) inflates the probability of at least one false alarm
- Hoga & Demetrescu (2023): monitoring statistics for VaR and ES with boundary functions
  - the boundary functions keep the probability of any false alarm over the whole horizon below the nominal level
- Compared with the traffic light: evidence accumulates from the start of monitoring
  - instead of a single 250-day snapshot
- The ES regression backtest (Bayer & Dimitriadis (2022)) of the case study can be monitored the same way
  - both address the one-shot design of the traffic light

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Conformal Prediction Beyond Exchangeability

- Barber, Candès, Ramdas & Tibshirani (2023): coverage without exchangeability
  - the coverage gap is bounded by weighted total-variation distances
  - between the score vector and the same vector with the test score swapped with calibration score i
- How large these distances are under volatility clustering is an empirical question
- Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of covariate shifts is attainable
  - which class captures a financial crisis is open
- Choosing the ACI step γ: Zaffran et al. (2022) aggregate ACI experts over a grid of γ values

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Case Study, Further Key Results

Finding	Source	Reading
Strict and Auxiliary ESR sizes differ by at most 0.01 for every process and sample size, including AR(1)-GARCH with $\phi = 0.5$	Table 1	In these designs, dropping the VaR forecasts barely changes the size; power is compared separately (Fig. 2–3)
Student-t, 250-day window: the 0.12 share comes from GARCH with a yearly refit; from 500 days GJR-GARCH-t and GARCH-t differ by at most 0.01	Table 3	Leverage in the variance equation barely changes the rejection shares once the window is long

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Case Study, Limitations and Extensions

- Limitation stated by the authors: auxiliary and strict ESR are two-sided only (the paper, Sec. 1.3)
- Limitation stated by the authors: the classical covariance over-rejects in small samples
- Replication (Seminar 8, C3): apply strict ESR (two-sided) and intercept ESR (one-sided)
  - to the ES 2.5% forecasts of the six models on the four markets
  - compare with Z_2 and McNeil–Frey
- Extension: replicate Table 3 with GJR-GARCH, more stocks and the period after August 2019
- Open question: the size of the two-sided intercept ESR at $T = 250$ (Table 1, EGARCH-t)
  - 0.04 with the robust covariance
  - 0.15 with the classical covariance
- The one-sided size is reported only for $T = 2\,500$, 0.02 (Table 2): what is it over one year?
- Can a bootstrap of the t-statistic hold it at 5% without losing the power of Fig. 4?

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Scorecard Across Four Markets

- Each cell: breach rate of VaR 1% in % / Z_2 p-value / MCS p-value (FZ0)
  - $\leq$0.001: at the resolution of the 1 000 bootstrap replications

Model	S&P 500	BET	Bitcoin	EUR/RON
HS	1.67 / <0.001 / $\leq$0.001	1.17 / 0.134 / $\leq$0.001	0.77 / 0.863 / n/a	0.51 / 1.000 / n/a
Normal	2.70 / <0.001 / $\leq$0.001	1.90 / <0.001 / $\leq$0.001	1.74 / 0.001 / n/a	1.06 / 0.973 / n/a
Student-t	2.00 / 0.002 / $\leq$0.001	1.58 / 0.003 / $\leq$0.001	0.71 / 0.625 / n/a	0.85 / 0.920 / n/a
GARCH-t	1.77 / <0.001 / 0.010	1.46 / <0.001 / 0.096	n/a / n/a / n/a	n/a / n/a / n/a
FHS	1.23 / 0.209 / 1.000	1.20 / 0.236 / 1.000	n/a / n/a / n/a	n/a / n/a / n/a
GARCH-EVT	1.15 / 0.145 / 0.211	0.97 / 0.215 / 0.966	n/a / n/a / n/a	n/a / n/a / n/a

- n/a: withheld GARCH forecast, hence no MCS (slide on withheld forecasts)

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Answers to the Numeric Self-Checks (1/2)

1. $\hat{\pi} = 0.006$; $\ln L(\alpha) = 497 \ln 0.99 + 3 \ln 0.01 = -18.81$, $\ln L(\hat{\pi}) = 497 \ln 0.994 + 3 \ln 0.006 = -18.34$
   - $\mathrm{LR}_{uc} = 2(-18.34 - (-18.81)) = 0.94 < 3.84$, $p = 0.33$: not rejected; 5 breaches were expected
2. $L = 3\%$: $\frac{3-2}{0.05 \cdot 2.5} + \frac{2}{2.5} + \ln 2.5 - 1 = 8 + 0.8 + 0.916 - 1 = 8.716$; $L = 1\%$: $0.8 + 0.916 - 1 = 0.716$
   - in decimals only $\ln e$ changes, by $\ln 0.01 = -4.605$: 4.111 and -3.889; the difference between the two days is the same

Self-checks

Answers to the Numeric Self-Checks (2/2)

3. $\lambda = 1000 \cdot 0.01^{\mathbf{2}}/(0.01 \cdot 0.99) = 10.10$, $\sqrt{\lambda} = 3.178$
   - power $= \Phi(1.218) + \Phi(-5.138) = 0.888$ (the second term is below $10^{-\mathbf{6}}$): the 88.8% of the dashed $T = 1000$ curve; exact power 78%
4. $\sum_t L_t I_t / \mathrm{ES}_t = 5.5$, $N_T = 5$, $T\alpha = 6.25$
   - $Z_{\mathbf{1}} = 1 - 5.5/5 = -0.10$: on breach days losses are on average 10% deeper than the ES forecast
   - $Z_{\mathbf{2}} = 1 - 5.5/6.25 = 0.12 > 0$: fewer breaches than the 6.25 expected offset that depth; far from the 5% threshold -0.70

Self-checks

The Discovery Loop (1/3): Prompts

- Prompts (tools: Semantic Scholar, Elicit, Google Scholar; check every DOI on Crossref)
  - `List peer-reviewed work since 2019 on conformal prediction under serial dependence, with DOI, exact guarantee and assumptions.`
  - `Give three rival explanations for crisis over-breaches of split conformal VaR and what each predicts for breach rates by regime.`
  - `Act as a hostile referee: list the strongest reasons this crisis effect is an artefact and the test that would settle each.`

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The Discovery Loop (2/3): Checks the Researcher Runs

- **Look-ahead bias**
  - change the observations after day $t-1$ and confirm that VaR_t, the scale and the crisis flag do not move
- **Dependent breaches**
  - Christoffersen and duration tests, or a block bootstrap of the hit sequence
- **Power**
  - 613 crisis days give 6.1 expected breaches of VaR 1%
  - a true 2% rate is detected with probability 57% by an exact two-sided Binomial test at 5% (i.i.d. breaches assumed)

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The Discovery Loop (3/3): Further Checks

- **Multiplicity**: 4 markets $\times$ 2 levels $\times$ 3 methods $=$ 24 tests
  - Holm controls the FWER (family-wise error rate)
  - Benjamini–Hochberg controls the FDR (false discovery rate) under independence or positive dependence (PRDS: positive regression dependency on a subset)
  - under arbitrary dependence: Benjamini–Yekutieli
- **Misstated theorems**
  - e.g. "ACI guarantees conditional coverage"
- **Off-by-one quantiles**
  - $k = \lceil (n+1)(1-\alpha) \rceil$ is not the default of `np.quantile`

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Project Seed, Crisis-Conditional Coverage

- Question: can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage?
- Question: does this coverage hold in all four markets under a pre-registered test?
- Replicate first
  - split conformal VaR 5% crisis rates (S&P 500 7.29%, EUR/RON 11.43%)
  - the ACI share of infinite VaR 1% (13.4% of S&P 500 evaluation days)
- Deliverable: a pre-registration (hypotheses, tests, corrections, power) dated before the first run
- Deliverable: a candidate method, e.g. Gibbs, Cherian & Candès (2025) with the volatility regime as conditioning class
  - VaR tests by regime
  - ES tested with the ES regression backtest (Bayer & Dimitriadis (2022)) or Z_2
- Deliverable: code and a two-page report with every specification tried
  - a referee-style critique of your own result
- Log every prompt that shaped a design choice in `AI_USE.md`
  - every error the assistant made in `AI_ERRORS.md`: github.com/danpele/MFM/tree/main/project

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Acronyms used in this material (1/3)

- **ACI**: Adaptive Conformal Inference
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model), as in AR(1)-GARCH
- **BCBS**: Basel Committee on Banking Supervision
- **BET**: Bucharest Exchange Trading (index)
- **BIS**: Bank for International Settlements
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **CC**: Conditional Coverage (test)
- **COVID-19**: COronaVIrus Disease 2019
- **DGP**: Data-Generating Process
- **DM**: Diebold–Mariano (test)
- **DOI**: Digital Object Identifier
- **DQ**: Dynamic Quantile (test)
- **EGARCH**: Exponential GARCH
- **ES**: Expected Shortfall
- **ESR**: Expected Shortfall Regression (backtest)

Acronyms used in this material (2/3)

- **EUR**: Euro
- **EVT**: Extreme Value Theory
- **FDR**: False Discovery Rate
- **FHS**: Filtered Historical Simulation
- **FRTB**: Fundamental Review of the Trading Book
- **FWER**: Family-Wise Error Rate
- **FZ**: Fissler–Ziegel (family of scoring functions for VaR and ES)
- **FZ0**: Fissler–Ziegel loss with zero-homogeneous score differences
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GARCH-EVT**: GARCH filter combined with Extreme Value Theory for the tail
- **GAS**: Generalised Autoregressive Score (model)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **GPD**: Generalised Pareto Distribution
- **GREE**: Growth Rate with Empirical e-statistics (e-backtest)
- **GREL**: Growth Rate with Empirical Losses (e-backtest)
- **GREM**: Growth Rate with a Mixture (of GREE and GREL)
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HS**: Historical Simulation

Acronyms used in this material (3/3)

- **IMA**: Internal Models Approach
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio
- **LRV**: Long-Run Variance
- **MCS**: Model Confidence Set
- **NASDAQ**: National Association of Securities Dealers Automated Quotations
- **OLS**: Ordinary Least Squares
- **P&L**: Profit and Loss
- **PIT**: Probability Integral Transform
- **POF**: Proportion Of Failures (test)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RON**: Romanian leu
- **S&P 500**: Standard & Poor's 500 (index)
- **VaR**: Value-at-Risk

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